

Mid

Quiz by Attiq Kundi

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CS-502

MIDTERM EXAMINATION

Fall 2011

CS502- Fundamentals of Algorithms

Question No: 1 (Marks: 1) - Please choose one

Due to left complete nature of binary tree, the heap can be stored in

- ▶ **Arrays (Page 40)**
- ▶ Structures
- ▶ Link Lis
- ▶ Stack

Question No: 1 (Marks: 1) - Please choose one

What type of instructions Random Access Machine (RAM) can execute?

- ▶ Algebraic and logic
- ▶ Geometric and arithmetic
- ▶ **Arithmetic and logic (Page 10)**
- ▶ Parallel and recursive

Question No: 1 (Marks: 1) - Please choose one

For Chain Matrix Multiplication we can not use divide and conquer approach because,

- ▶ **We do not know the optimum k (Page 86)**
- ▶ We use divide and conquer for sorting only
- ▶ We can easily perform it in linear time
- ▶ Size of data is not given

Question No: 1 (Marks: 1) - Please choose one

What is the total time to heapify?

- ▶ **$O(\log n)$ (Page 43)**
- ▶ $O(n \log n)$
- ▶ $O(n^2 \log n)$
- ▶ $O(\log^2 n)$

Question No: 1 (Marks: 1) - Please choose one

word Algorithm comes from the name of the muslim author _____

- ▶ **Abu Ja'far Mohammad ibn Musa al-Khowarizmi.**

Question No: 1 (Marks: 1) - Please choose one al-Khwarizmi's work was written in a book titled _____

- ▶ **al Kitab al-mukhatasar fi hisab al-jabr wa'l-muqabalah**

MIDTERM EXAMINATION

Spring 2010

CS502- Fundamentals of Algorithms

Question No: 1 (Marks: 1) - Please choose one

Random access machine or RAM is a/an

- ▶ Machine build by Al-Khwarizmi
- ▶ Mechanical machine
- ▶ Electronics machine
- ▶ **Mathematical model (Page 10)**

Question No: 2 (Marks: 1) - Please choose one

_____ is a graphical representation of an algorithm

- ▶ Σ notation
- ▶ Θ notation
- ▶ **Flowchart [Click here for detail](#)**
- ▶ Asymptotic notation

Question No: 3 (Marks: 1) - Please choose one

A RAM is an idealized machine with _____ random-access memory.

- ▶ 256MB
- ▶ 512MB
- ▶ **an infinitely large (Page 10)**
- ▶ 100GB

Question No: 4 (Marks: 1) - Please choose one

What type of instructions Random Access Machine (RAM) can execute? Choose best answer

- ▶ Algebraic and logic
- ▶ Geometric and arithmetic
- ▶ **Arithmetic and logic (Rep)**
- ▶ Parallel and recursive

Question No: 5 (Marks: 1) - Please choose one

What will be the total number of max comparisons if we run brute-force maxima algorithm with n elements?

- ▶ n^2
- ▶ n
- ▶ n^2
- ▶ n **(Page 14)**
- ▶ n^8

Question No: 6 (Marks: 1) - Please choose one

What is the solution to the recurrence $T(n) = T(n/2) + n$.

- ▶ $O(\log n)$
- ▶ **$O(n)$ (Page 37)**
- ▶ $O(n \log n)$
- ▶ $O(n^2)$

Question No: 7 (Marks: 1) - Please choose one

Consider the following code:

```
For(j=1; j<n; j++)
    For(k=1; k<15; k++)
        For(l=5; l<n; l++)
            {
```

```
Do_something_constant();
```

```
}
```

What is the order of execution for this code.

- ▶ $O(n^3)$
- ▶ $O(n^2)$
- ▶ $O(n^2 \log n)$
- ▶ $O(n^2)$

Question No: 8 (Marks: 1) - Please choose one

What is the total time to heapify?

- ▶ $O(\log n)$ rep
- ▶ $O(n \log n)$
- ▶ $O(n^2 \log n)$
- ▶ $O(\log^2 n)$

Question No: 9 (Marks: 1) - Please choose one

Consider the following Algorithm:

Factorial (n){

if (n=1)

return 1

else

return (n * Factorial(n-1))

Recurrence for the following algorithm is:

- ▶ $T(n) = T(n-1) + 1$
- ▶ $T(n) = nT(n-1) + 1$
- ▶ $T(n) = T(n-1) + n$
- ▶ $T(n) = T(n(n-1)) + 1$

Question No: 10 (Marks: 1) - Please choose one

When we call heapify then at each level the comparison performed takes time

► **It will take $\Theta(1)$ (Page 43)**

- Time will vary according to the nature of input data
- It can not be predicted
- It will take $\Theta(\log n)$

Question No: 11 (Marks: 1) - Please choose one

In Quick sort, we don't have the control over the sizes of recursive calls

► **True(Page 40)**

- False
- Less information to decide
- Either true or false

Question No: 12 (Marks: 1) - Please choose one Is it possible to sort without making comparisons?

► **Yes(Page 57)**

- No

Question No: 13 (Marks: 1) - Please choose one

If there are $\Theta(n^2)$ entries in edit distance matrix then the total running time is ► $\Theta(1)$

► **$\Theta(n^2)$ [Click here for detail](#)**

- $\Theta(n)$
- $\Theta(n \log n)$

Question No: 14 (Marks: 1) - Please choose one

For Chain Matrix Multiplication we can not use divide and conquer approach because,

► **We do not know the optimum k (Page 86)**

- We use divide and conquer for sorting only
- We can easily perform it in linear time

- ▶ Size of data is not given

Question No: 15 (Marks: 1) - Please choose one

The Knapsack problem belongs to the domain of _____ problems.

▶ **Optimization (Page 91)**

- ▶ NP Complete
- ▶ Linear Solution
- ▶ Sorting

Question No: 16 (Marks: 1) - Please choose one

Suppose we have three items as shown in the following table, and suppose the capacity of the knapsack is 50 i.e. $W = 50$.

Item	Value	Weight
1	60	10
2	100	20
3	120	30

The optimal solution is to pick

- ▶ Items 1 and 2
- ▶ Items 1 and 3
- ▶ **Items 2 and 3 (correct)**
- ▶ None of these

MIDTERM EXAMINATION
Spring 2010
CS502- Fundamentals of Algorithms

Question No: 1 (Marks: 1) - Please choose one

For the Sieve Technique we take time

▶ **$T(nk)$ (Page 34)**

▶ $T(n / 3)$

▶ n^2

▶ $n/3$

Question No: 1 (Marks: 1) - Please choose one

Sieve Technique applies to problems where we are interested in finding a single item

from a larger set of _____

Select correct option:

▶ **n items (Page 34)**

▶ phases

▶ pointers

▶ constant

Question No: 1 (Marks: 1) - Please choose one

_____ graphical representation of algorithm.

▶ asymptotic

► **Flowchart (rep)**

Question No: 1 (Marks: 1) - Please choose one

who invented the quick sort

► **C.A.R. Hoare** [Click here for detail](#)

Question No: 1 (Marks: 1) - Please choose one

main elements to a divide-and-conquer

► **Divide, conquer, combine (Page 27)**

Question No: 1 (Marks: 1) - Please choose one Mergesort
is a stable algorithm but not an in-place algorithm.

► **True (Page 54)**

► false

Question No: 1 (Marks: 1) - Please choose one

Counting sort the numbers to be sorted are in the range 1 to k where k is small.

► **True (Page 57)**

► False

MIDTERM EXAMINATION
Spring 2007
CS502- Fundamentals of Algorithms

Question No: 1 (Marks: 1) - Please choose one

Total time for heapify is:

- ▶ $O(\log^2 n)$
- ▶ $O(n \log n)$
- ▶ $O(n^2 \log n)$
- ▶ **$O(\log n)$**

Rep

Question No: 1 (Marks: 1) - Please choose one

If an algorithm has a complexity of $\log_2 n + n \log_2 n + n$. we could say that it has complexity

- ▶ $O(n)$
- ▶ $O(n \log_2 n)$
- ▶ $O(3)$
- ▶ $O(\log_2 (\log_2 n))$
- ▶ $O(\log_2 n)$

Question No: 1 (Marks: 1) - Please choose one In
RAM model instructions are executed

- ▶ **One after another (Page 10)**

► Parallel

► Concurrent

► Random

Question No: 1 (Marks: 1) - Please choose one

In selection algorithm, because we eliminate a constant fraction of the array with each phase, we get the

▶ **Convergent geometric series (Page 37)**

▶ Divergent geometric series

▶ None of these

Question No: 1 (Marks: 1) - Please choose one

Due to left-complete nature of binary tree, heaps can be stored in

▶ Link list

▶ Structure

▶ **Array (Page 40)**

▶ None of above

CS609- System Programming

Midterm Quizzes (Quiz No.1 & 2)

Quiz No.1 (04 – MAY - 2013)

Question No: 1 (Marks: 1) - Please choose one

The time assumed for each basic operation to execute on RAM model of computation is-----

Infinite

Continuous

Constant (Page 10)

Variable

Question No: 1 (Marks: 1) - Please choose one

If the indices passed to merge sort algorithm are not equal, the algorithm may return immediately.

True

False (Page 28)

Question No: 1 (Marks: 1) - Please choose one Brute-force algorithm uses no intelligence in pruning out decisions.

True (Page 18)

False

Question No: 1 (Marks: 1) - Please choose one

In analysis, the Upper Bound means the function grows asymptotically no faster than its largest term.

True (Page 24)

False

Question No: 1 (Marks: 1) - Please choose one

For small values of n , any algorithm is fast enough. Running time does become an issue when n gets large.

True (Page 14)

Fast

Question No: 1 (Marks: 1) - Please choose one

The array to be sorted is not passed as argument to the merge sort algorithm.

True

False

Question No: 1 (Marks: 1) - Please choose one

In simple brute-force algorithm, we give no thought to efficiency.

True
False (Page 11)

Question No: 1 (Marks: 1) - Please choose one

The ancient Roman politicians understood an important principle of good algorithm design that is plan-sweep algorithm.

True

False (Page 27) [Divide and Conquer]

Question No: 1 (Marks: 1) - Please choose one

In 2d-space a point is said to be _____if it is not dominated by any other point in that space.

Member

Minimal

Maximal (Page 11)

Joint

Question No: 1 (Marks: 1) - Please choose one

An algorithm is a mathematical entity that is dependent on a specific programming language.

True **False**

Question No: 1 (Marks: 1) - Please choose one

The running time of an algorithm would not depend upon the optimization by the compiler but that of an implementation of the algorithm would depend on it.

True (Page 13)

False

Question No: 1 (Marks: 1) - Please choose one

$f(n)$ and $g(n)$ are asymptotically equivalent. This means that they have essentially the same _____ for large n .

Results

Variables

Size

Growth rates (Page 23)

Question No: 1 (Marks: 1) - Please choose one

$8n^2 + 2n - 3$ will eventually exceed $c_2 \cdot n$ no matter how large we make c_2 .

True (Page 25)

False

Question No: 1 (Marks: 1) - Please choose one

If we associate (x, y) integers pair to cars where x is the speed of the car and y is the negation of the price. High y value for a car means a _____ car.

Fast

Slow

Expensive

Cheap **(Page 11)**

Question No: 1 (Marks: 1) - Please choose one

The function $f(n) = n(\log n + 1)/2$ is asymptotically equivalent to $n \log n$. Here Upper Bound means the function $f(n)$ grows asymptotically _____ faster than $n \log n$.

More

Quiet

Not (Page 24)

At least

Question No: 1 (Marks: 1) - Please choose one

After sorting in merge sort algorithm, merging process is invoked.

Select correct option:

True (Page 28)

False

Question No: 1 (Marks: 1) - Please choose one

Asymptotic growth rate of the function is taken over _____ case running time.

Select correct option:

Best

Average

Worst (Page 14)

Normal

Question No: 1 (Marks: 1) - Please choose one

In analysis of $f(n) = n(n/5) + n - 10 \log n$, $f(n)$ is asymptotically equivalent to _____.

n

$2n$

$n+1$

n^2 (Page 23)

Question No: 1 (Marks: 1) - Please choose one

Algorithm is concerned with.....issues.

Macro

Micro

Both Macro & Micro (Page 8)

Normal

Question No: 1 (Marks: 1) - Please choose one

We cannot make any significant improvement in the running time which is better than that of brute-force algorithm.

True

False (Page 18)

Question No: 1 (Marks: 1) - Please choose one

In addition to passing in the array itself to Merge Sort algorithm, we will pass in _____ other arguments which are indices.

Two (Page 28)

Three

Four

Five

Question No: 1 (Marks: 1) - Please choose one

Consider the following Algorithm: Fun(n){ if (n=1) return 1 else return (n * Fun(n-1)) } Recurrence for the above algorithm is:

$nT(n-1)+1$

$2T(n-1)+1$

$T(n-1)+cn$

$T(n-1)+1$

Question No: 1 (Marks: 1) - Please choose one

In analysis, the Lower Bound means the function grows asymptotically at least as fast as its largest term.

True (Page 24)

False

Question No: 1 (Marks: 1) - Please choose one

Efficient algorithm requires less computational.....

Memory

Running Time

Memory and Running Time (Page 9)

Energy

Question No: 1 (Marks: 1) - Please choose one

The O-notation is used to state only the asymptotic _____ bounds.

Two

Lower

Upper (Page 25)

Both lower & upper

Question No: 1 (Marks: 1) - Please choose one

For the worst-case running time analysis, the nested loop structure containing one “for” and one “while” loop, might be expressed as a pair of _____ nested summations.

1

2 (Page 16)

3

4

Question No: 1 (Marks: 1) - Please choose one

Before sweeping a vertical line in plane sweep approach, in start sorting of the points is done in increasing order of their _____ coordinates.

X (Page 18)

Y

Z

X & Y

Question No: 1 (Marks: 1) - Please choose one

Brute-force algorithm for 2D-Maxima is operated by comparing _____ pairs of points.

Two

Some

Most

All (Page 18)

Question No: 1 (Marks: 1) - Please choose one

The function $f(n) = n(\log n + 1)/2$ is asymptotically equivalent to $n \log n$. Here Lower Bound means function $f(n)$ grows asymptotically at _____ as fast as $n \log n$.

Normal

Least (Page 23)

Most

All

Question No: 1 (Marks: 1) - Please choose one

The definition of Theta-notation relies on proving _____ asymptotic bound.

One

Lower

Upper

Both lower & upper (Page 25) rep

Question No: 1 (Marks: 1) - Please choose one

In plane sweep approach, a vertical line is swept across the 2d-plane and _____ structure is used for holding the maximal points lying to the left of the sweep line.

Array

Queue

Stack (Page 18)

Tree

Question No: 1 (Marks: 1) - Please choose one

Algorithm analysts know for sure about efficient solutions for NP-complete problems.

Select correct option:

True

False (Page 9)

Quiz No.1 (2012)

Question No: 1 of 10 (Marks: 1) - Please choose one The number of nodes in a complete binary tree of height h is

$2^{(h+1)} - 1$ (Page 40)

$2 * (h+1) - 1$

$2 * (h+1)$

$((h+1) ^ 2) - 1$

Question No: 1 of 10 (Marks: 1) - Please choose one

The analysis of Selection algorithm shows the total running time is indeed _____in n,

arithmetic

geometric

linear (Page 37)

orthogonal

Question No: 1 of 10 (Marks: 1) - Please choose one

A (an) _____ is a left-complete binary tree that conforms to the heap order

heap (Page 40)

binary tree

binary search tree

array

Question No: 1 of 10 (Marks: 1) - Please choose one Analysis of Selection algorithm ends up with,

T(n) (Page 37)

$T(1 / 1 + n)$

$T(n / 2)$

$T((n / 2) + n)$

Question No: 1 of 10 (Marks: 1) - Please choose one For the sieve technique we solve the problem,

recursively (Page 34)

mathematically

precisely

accurately

Question No: 1 of 10 (Marks: 1) - Please choose one

A heap is a left-complete binary tree that conforms to the _____

increasing order only

decreasing order only

heap order (Page 40)

(log n) order

Question No: 1 of 10 (Marks: 1) - Please choose one In which order we can sort?

increasing order only

decreasing order only

increasing order or decreasing order (Page 39)

both at the same time

Question No: 1 of 10 (Marks: 1) - Please choose one Divide-and-conquer as breaking the problem into a small number of

pivot

Sieve

smaller sub problems (Page 34)

Selection

Question No: 1 of 10 (Marks: 1) - Please choose one For the heap sort we store the tree nodes in

level-order traversal (Page 40)

in-order traversal

pre-order traversal

post-order traversal

Question No: 1 of 10 (Marks: 1) - Please choose one

The sieve technique works in _____ as follows

Phases (Page 34)

numbers

integers

routines

CS502 - Fundamentals of Algorithms

Quiz No.1 12-11-2012

Question No: 1 of 10 (Marks: 1) - Please choose one

We do sorting to,

keep elements in random positions

keep the algorithm run in linear order

keep the algorithm run in $(\log n)$ order

keep elements in increasing or decreasing order

Question No: 1 of 10 (Marks: 1) - Please choose one

Heaps can be stored in arrays without using any pointers; this is due to the _____ nature of the binary tree,

left-complete(Page 40)

right-complete

tree nodes

tree leaves

Question No: 1 of 10 (Marks: 1) - Please choose one

Sieve Technique can be applied to selection problem?

True (Page 35)

False

Question No: 1 of 10 (Marks: 1) - Please choose one

In Sieve Technique we do not know which item is of interest

True (Page 34)

False

Question No: 1 of 10 (Marks: 1) - Please choose one

In the analysis of Selection algorithm, we eliminate a constant fraction of the array with each phase; we get the convergent _____ series in the analysis,

linear

arithmetic

geometric (Page 37)

exponent

Question No: 1 of 10 (Marks: 1) - Please choose one

For the heap sort, access to nodes involves simple _____ operations.

arithmetic (Page 41)

binary

algebraic

logarithmic

Question No: 1 of 10 (Marks: 1) - Please choose one Slow sorting algorithms run in,

$T(n^2)$ (Page 39)

$T(n)$

$T(\log n)$

Question No: 1 of 10 (Marks: 1) - Please choose one

In the analysis of Selection algorithm, we make a number of passes, in fact it could be as many as,

$T(n)$

$T(n / 2)$

$\log n$ (Page 37)

$n / 2 + n / 4$

Question No: 1 of 10 (Marks: 1) - Please choose one

The sieve technique is a special case, where the number of sub problems is just

5

many

1 (Page 34)

few

Question No: 1 of 10 (Marks: 1) - Please choose one

How many elements do we eliminate in each time for the Analysis of Selection algorithm?

$(n / 2) + n$ elements

$(n / 2)$ elements (Page 37)

$n / 4$ elements

$2n$ elements

Question No: 1 of 10 (Marks: 1) - Please choose one

One of the clever aspects of heaps is that they can be stored in arrays without using any _____.

pointers (Page 40)

constants

variables

functions

Question No: 1 of 10 (Marks: 1) - Please choose one

How much time merge sort takes for an array of numbers?

$T(n^2)$

$T(n)$

$T(\log n)$

$T(n \log n)$ (Page 40)

Question No: 1 of 10 (Marks: 1) - Please choose one

The reason for introducing Sieve Technique algorithm is that it illustrates a very important special case of,

divide-and-conquer (Page 34)

decrease and conquer

greedy nature

2-dimension Maxima

Question No: 1 of 10 (Marks: 1) - Please choose one In Sieve Technique we do not know which item is of interest

True (Page 34) rep

False

Question No: 1 of 10 (Marks: 1) - Please choose one

Theta asymptotic notation for $T(n)$:

Set of functions described by: $c_1g(n)$ Set of functions described by $c_1g(n) \geq f(n)$ for c_1 s

Theta for $T(n)$ is actually upper and worst case comp (Not sure)

Set of functions described by: $c_1g(n)$

Question No: 1 of 10 (Marks: 1) - Please choose one

Memoization is?

To store previous results for future use

To avoid this unnecessary repetitions by writing down the results of recursive calls and looking them up again if we need them later (page 74)

To make the process accurate

None of the above

Question No: 1 of 10 (Marks: 1) - Please choose one

Which sorting algorithm is faster

$O(n \log n)$ Page 26

$O(n^2)$

$O(n^2)$

$O(n^2)$

$O(n^2)$

$O(n^3)$

Question No: 1 of 10 (Marks: 1) - Please choose one Quick sort is

Stable & in place

Not stable but in place (Page 54)

Stable but not in place

Some time stable & some times in place

Question No: 1 of 10 (Marks: 1) - Please choose one One example of in place but not stable algorithm is

Merger Sort

Quick Sort (Page 54)

Continuation Sort

Bubble Sort

Question No: 1 of 10 (Marks: 1) - Please choose one

Cont sort is suitable to sort the elements in range 1 to k

K is Large

K is not known

K may be small or large

K is small (Page 57)

Question No: 1 of 10 (Marks: 1) - Please choose one

In place stable sorting algorithm.

If duplicate elements remain in the same relative position after sorting (Page 54)

One array is used

More than one arrays are required

Duplicating elements not handled

Question No: 1 of 10 (Marks: 1) - Please choose one

Which may be a stable sort?

Merger

Insertion (Page 54)

Both above

None of the above

Question No: 1 of 10 (Marks: 1) - Please choose one

An in place sorting algorithm is one that uses ____ arrays for storage

Two dimensional arrays

More than one array

No Additional Array (Page 54)

None of the above

Question No: 1 of 10 (Marks: 1) - Please choose one

Sieve Technique applies to problems where we are interested in finding a single item from a larger set of _____

n items (Page 34)

phases

pointers

constant

Question No: 1 of 10 (Marks: 1) - Please choose one

Sorting is one of the few problems where provable _____ bounds exist on how fast we can sort,

upper

lower (Page 39)

average

$\log n$

Question No: 1 of 10 (Marks: 1) - Please choose one

Counting sort has time complexity:

$O(n)$ (Page 58)

$O(n+k)$

$O(k)$

$O(n \log n)$

Question No: 1 of 10 (Marks: 1) - Please choose one

The running time of quick sort depends heavily on the selection of

No of inputs

Arrangement of elements in array

Size o elements

Pivot elements (Page 49)

Question No: 1 of 10 (Marks: 1) - Please choose one

Which may be stable sort:

Bubble sort

Insertion sort

Both of above (Page 54)

Question No: 1 of 10 (Marks: 1) - Please choose one

One Example of in place but not stable sort is

Quick (Page 54)

Heap

Merge

Bubble

Question No: 1 of 10 (Marks: 1) - Please choose one

In Quick Sort Constants hidden in $T(n \log n)$ are

Large

Medium

Small [Click here for detail](#)

Not Known

Question No: 1 of 10 (Marks: 1) - Please choose one

Quick sort is based on divide and conquer paradigm; we divide the problem on base of pivot element and:

There is explicit combine process as well to conquer the solution.

No work is needed to combine the sub-arrays, the array is already sorted

Merging the sub arrays

None of above. (Page 51)

Ref: - random choices for the pivot element and each choice have an equal probability of $1/n$ of occurring.
So we can modify the above recurrence to compute an average rather than a max

CS501 - Quiz No.2 (Spring 2013)

Question No: 1 of 10 (Marks: 1) - Please choose one

A point p in 2-dimensional space is usually given by its integer coordinate(s)_____

p.x only

p.y only

p.x & p.z

p.x & p.y (Page 10)

Question No: 1 of 10 (Marks: 1) - Please choose one

In _____ we have to find rank of an element from given input.

Merge sort algorithm

Selection problem (Page 34)

Brute force technique

Plane Sweep algorithm

Question No: 1 of 10 (Marks: 1) - Please choose one

In Heap Sort algorithm, if heap property is violated _____

We call Build heap procedure

We call Heapify procedure

We ignore

Heap property can never be violated

Question No: 1 of 10 (Marks: 1) - Please choose one

Upper bound requires that there exist positive constants c_2 and n_0 such that $f(n) \leq c_2 n$ for all $n \geq n_0$ (yes)

question ghalat lag raha hai mujhae

Less than

Equal to or Less (Page 25)
than Equal or Greater
than Greater than

Question No: 1 of 10 (Marks: 1) - Please choose one

A RAM is an idealized algorithm with takes an infinitely large random-access memory.

True

False (Page 10)

Question No: 1 of 10 (Marks: 1) - Please choose one

_____ is one of the few problems, where provable lower bounds exist on how fast we can sort.

Searching

Sorting (Page)

Both Searching & Sorting

Graphing

Question No: 1 of 10 (Marks: 1) - Please choose one

Floor and ceiling are _____ to calculate while analyzing algorithms.

Very easy

Usually considered difficult (Page 31)

Question No: 1 of 10 (Marks: 1) - Please choose one

In Heap Sort algorithm, the maximum levels an element can move upward is _____

Theta ($\log n$) (Page 43)

Order ($\log n$)

Omega ($\log n$)

$O(1)$ i.e. Constant time

Question No: 1 of 10 (Marks: 1) - Please choose one

A point p in 2-dimensional space is usually given by its integer coordinate(s)_____

$p.x$ only $p.y$

only $p.x$ & $p.z$

$p.x$ & $p.y$ (Page 17)

Question No: 1 of 10 (Marks: 1) - Please choose one

In Heap Sort algorithm, the total running time for Heapify procedure is _____

Theta ($\log n$) (Page 43)

Order ($\log n$)

Omega ($\log n$)

$O(1)$ i.e. Constant time

Question No: 1 of 10 (Marks: 1) - Please choose one

Algorithm is a mathematical entity, which is independent of a specific machine and operating system.

True

False (Page 7)

Question No: 1 of 10 (Marks: 1) - Please choose one

While Sorting, the ordered domain means for any two input elements x and y _____ satisfies only.

$x < y$

$x > y$

$x = y$

All of the above (Page 39)

Question No: 1 of 10 (Marks: 1) - Please choose one Quick sort is best from the perspective of Locality of reference.

True
False (Page 9)

Question No: 1 of 10 (Marks: 1) - Please choose one

Sorting can be in _____

Increasing order only

Decreasing order only

Both Increasing and Decreasing order (Page 39)

Random order

Question No: 1 of 10 (Marks: 1) - Please choose one

In Heap Sort algorithm, we build _____ for ascending sort.

Max heap
Min heap (Page 41)

Question No: 1 of 10 (Marks: 1) - Please choose one In Sieve Technique, we know the item of interest.

True

False (Page 34)

Question No: 1 of 10 (Marks: 1) - Please choose one

While solving Selection problem, in Sieve technique we partition input data _____

In increasing order

In decreasing order

According to Pivot (Page 3

Randomly

Question No: 1 of 10 (Marks: 1) - Please choose one

In pseudo code, the level of details depends on intended audience of the algorithm.

True (Page 12)

False

Question No: 1 of 10 (Marks: 1) - Please choose one

The sieve technique works where we have to find _____ item(s) from a large input.

Single (Page 34)

Two

Three

Similar

Question No: 1 of 10 (Marks: 1) - Please choose one

If the indices passed to merge sort algorithm are _____, then this means that there is only one element to sort.

Small

Large

Equal (Page 28)

Not Equal

1) The sieve technique works where we have to find ----- item(s) from a large input.

- **Single** **page 34**

- Two
- Three
- Similar

2) The only way to convert an empty string into a string of j characters is by doing j insertions, represented as

- $E(i, j) = I$
- $E(i, 0) = i$
- **$E(0, j) = j$** **page 78**
- $E(1, j) = j$

3) In Selection problem, the Sieve technique works in -----

- Non-recursive manner
- Constant time
- **Phases** **page 34**
- One complete go

4) Algorithm is a sequence of computational steps that ----- the input into output.

- Merge
- Assign
- **Transform** **page 7**
- Integrate

5) In max heap (for Heap Sort algorithm), when every time maximum element is removed from top we replace it with ----- in the leaf tree.

- Second last
- **Last** **page 41**
- First
- Any

6) If p_j dominates p_i and p_i dominates p_h then p_j also dominates p_h . It means dominance relation is

- **Transitive** page 18
- Non transitive
- Equation

Symbolic

7) In the following code the statement “cout<<j;” executes ----- times

```
For(j=1;j<=5;j=j+2)
```

```
Cout<<j;
```

- 5 times
- 2 times
- **3 times (conceptual)**
- 0 times

9) For average-case time analysis of Quick sort algorithm, Pivot selection is on average basis from -----.

- Half of the input values
- **All possible random values** page 50
- Pivot is input separately
- Values greater than 5

10) In the statement, “output $P[i].x, P[i].y$ ”, the number of times elements of P are accessed is -----.

- 1
- **2** page 14
- 3
- 4

11) Identify the TRUE statement.

- The Knapsack problem does not belong to the domain of optimization problems.
- **The Knapsack problem belongs to the domain of optimization problems** Page 91
- The Knapsack problem cannot be solved by using dynamic programming.
- The Knapsack problem is optimally solved by using brute force algorithm.

12) Matrix multiplication is a(n) ----- operation.

- Commutative
- **Associative page 85**
- Neither commutative nor associative
- Commutative but not associative

13) ----- provides us more accurate result when input values are not closer to each other.

- Average

• **Median page 34(conceptual)**

14) In merge sort algorithm, we split the array around the ----- index q.

- Entering
- **Mid page 28**
- Exitin

Summing

14) In the analysis of Selection algorithm, we get the convergent ----- series.

- Harmonic
- Linear
- Arithmetic
- **Geometric page 37**

16) In Selection problem, the Sieve technique -----.

- Adds some more input items each time.
- Do not work recursively
- Do not uses Divide and Conquer approach
- **Eliminate undesired data items each time page 35**

17) If matrix A of dimension $p \times q$ is multiply with matrix B of dimension $q \times r$, then each entry in resultant matrix takes ----- time.

- **$O(q)$ page 84**

- $O(1)$
- $O(p \times q)$
- $O(q \times r)$

18) In Sorting the key value or attribute ----- form an ordered domain.

- Must be

Not always **page 39**

19) In Heap Sort algorithm, the total running time for Heapify procedure is ----- .

Theta
(log n) **page**
43

- Order (log n)
- Omega (log n)
- $O(1)$ i.e. Constant time

20) Pseudo code of algorithm are to be read by -----.

- **People** **page 12**
- RAM
- Computer
- Compiler

21) In Dynamic Programming based solution of Knapsack Problem, if we decide to take an object 'i', then we gain -----.

- W (Total Weight of Knapsack)
- V (Total Value of all items)
- **v_i (Value of object i)** **page 93**
- None of the given option

22) The worst case running time of Quick sort algorithm -----.

- Cannot be quadratic

Is quadratic (google)

- Is always exponential
- Is linear

23) The only way to convert a string of i characters into the empty string is with i deletions , represented as

- $E(0, j)=j$
- $E(i, j)=l$
- $E(0, i)=j$

- **$E(i, 0)=i$ page 78**

24) In Selection problem, the rank of an element will be its ----- position if we sort the input data.

- First
- **Final page 34**
- Second last
- Last

25) Which one is not passed as parameter in Quick sort algorithm?

- End of array
- Middle of array

• Array (containing input elements) (cs301)

- Start of the array

26) While Sorting, the ordered domain means for any two input elements x and y ----- satisfies only.

- $X < y$
- $X > y$

- $X=y$
- **All of the above** **page 39**

27) Which of the following is calculated with Big O notation?

- Medium bounds
- **Upper bounds** **page 25**
- Lower bounds
- Both upper and lower bounds

28) For solving Selection problem, we introduced Sieve technique due to -----

- Using Decrease and Conquer strategy
- Avoiding to sort all input data
- **Eliminating Rank of an element** **page 35**

Using Brute-Force approach

29) The worst-case running time of merge sort is ----- in order to sort an array of n elements.

- $O(\log n)$
- $O(n)$

$O(n \log n)$ page 30

-

30) Quick sort is a recursive algorithm.

- False
- **True (page 49)**

31) While analyzing Selection algorithm, we make a number of passes , in fact it could be as many as -----.

- $n(n+1)$
- **$\log(n)$ page 37**

- $n/3$
- $n/4$

32) Approach for solving geometric problems by sweeping a line across the plane is called ----- sweep.

- Line
- **Plane** **page 18**
- Cube
- Box

33) ----- time is the maximum running time over all legal inputs.

- **Worst-case** **page 13**
- Average-case
- Best-case
- Good-case

34) The asymptotic growth of $n(n+1)/2$ is:

- **$O(2)$** **page 15**
- $O(n)$
- $O(n+2)$
- $O(n \log n)$

35) One of the limitation in 0/1 Knapsack is that an item can either be ----- in the bag or not.

- Use
- **Put** **page 91**
- Move
- Store

36) ----- is one of the few problems, where provable lower bounds exist on how fast we can sort.

- Searching
- **Sorting** **page 39**
- Both searching and sorting
- Graphing

37) Selection algorithm takes theta -----.

- (n^2)
- **(n) page 37**
- $\log(n)$
- $n\log(n)$

38) The Sieve technique is a special case, where the numbers of sub-problems is just -----.

- **1 page 34**
- 2
- 3
- 4

39) Following is not the application of Edit Distance problem

- Speech recognition
- Spelling Correction
- **Ascending Sort page 76**
- Computational Molecular Biology

40) We can use the optimal substructure property to devise a ----- formulation of the edit distance problem.

- Selective
- Optimum
- Iterative

Recursive (p78)

41) In plane sweep approach, a vertical line is swept across the 2d-plane form -----

- Right to left
- **Left to right** **page 18**
- Top to bottom
- Bottom to top

42) In 3-Dimensional space, a point P has ----- coordinate(s).

- (X,Y)
- (X,0)
- (0,Y)
- **(X,Y,Z)** **(google)**

43) ----- items are not allowed in the 0/1 Knapsack.

- Lighter
- **Fractional** **page 91**
- Whole
- Weighty

44) Radix sort is not a non-comparative integer sorting algorithm.

- True
- **False** **(google)**

45) Fibonacci sequence was named on -----, a famous mathematician in 12th Century.

- Fred Brooks
- Grady Booch
- **Leonardo Pisano** **page 73**
- Edgar F.Codd

46) In generating Fibonacci Sequence, we can avoid unnecessary repetitions by ----- process.

- Tokenization
- **Memoization page 74**
- Randomization
- Memorization

47) The sequence of merge sort algorithm is:

- Divide-Combine-Conquer
- Conquer-Divide-Combine
- **Divide-Conquer-Combine page 27**
- Combine-Divide-Conquer

48) Sorting can be in -----.

- Increasing order only
- Decreasing order only
- **Both Increasing and Decreasing order page 52**
- Random order

49) The process of ----- ends when you are left with such tiny pieces remaining that it is trivial to solve them.

- Brute-force
- Plan-sweep
- **Divide and Conquer page 27**
- Axis-sweep

50) Time complexity of the “0-1” knapsack algorithm depends on -----.

- Number of items
- Capacity of the knapsack
- Size of the table
- **Number of items and capacity of knapsack (Google)**

51) In max-heap, largest element is stored at root node. Where is the smallest element stored?

- Right Node

Leaf Node (Google)

- Middle Node
- Left Node

52) For ----- values of n, any-----algorithm is fast enough.

Small pg#14

- Medium
- Large
- Infinity

53) To predict the cost of an algorithm in terms of resources is called -----.

- Key

Analysis (Google)

- Design
- Output

54) Dynamic programming comprises of -----

• Recursion only pg#75

- Repetition only
- Recursion and Repetition
- No Repetition but Recursion

55) For Quick sort algorithm, Partitioning takes theta -----?

- (n)
- $\log(n)$
- **$n\log(n)$ (Google)**
- $n^2\log(n)$

56) An in-place sorting algorithm is one that ----- uses additional array for storage.

- Always
- Permanently
- **Does not pg#54**
- Sometime

57) In Dynamic Programming approach, we do not store the solution to each sub-problem in case if it reappears.

- True
- **False pg#75**

58) Counting sort has time complexity:

- **$O(n+k)$ pg#58**
- $O(n)$
- $O(k)$
- $O(n\log n)$

59) Due to left complete nature of binary tree, the heap can be stored in

- **Arrays pg#40**
- Structures
- Link list
- Stack

60) For the heap sort we store the tree node in

- **Level-order traversal pg#40**

- In-order traversal
- Pre-order traversal
- Post-order traversal

61) In the clique cover problem, for two vertices to be in the same group, they must be -----each other.

- Apart from
- Far from
- Near to

• **Adjacent to pg#176**

62) In in-place sorting algorithm is one that uses arrays for storage :

No additional array pg#54

- An additional array
- Both of above may be true according to algorithm
- More than 3 arrays of one dimension

63) Brute-force algorithm for 2D-Maxima is operated by comparing ----- pairs of points.

- Two
- Some
- Most

All pg#18

64) $F(n)$ and $g(n)$ are asymptotically equivalent. This means they have essentially the same ----- for large n .

- Results

Growth rates pg#23

- Variables
- Size

65) Quick sort is

- Stable & in place
- **Not stable but in place pg#54**
- Stable but not in place
- Some time stable & some times in place

66) Heaps can be sorted in arrays without using any pointers; this is due to the ----- nature of the binary tree.

- **Left-complete pg#40**

- Right-complete
- Tree nodes
- Tree leaves

67) Which may be a stable sort?

- **Merger**

- **Insertion**

- **Both above page 54**
- None of the above

68) For the Sieve Technique we take time

- **$T(nk)$ page 34**
- $IT(n/3)$
- n^2
- $n/3$

69) One of clever aspects of heaps is that they can be stored in arrays without using any -----.

- **Pointers page 40**
- Constants
- Variables
- Functions

70) Counting sort is suitable to sort the elements in range 1 to k.

- K is large
- K is not known
- K may be small or large
- **K is small page 57**

71) Asymptotic growth rate of the function is taken over ----- case running time.

- Best
- **Worst page 14**
- Average
- Normal

72) The Sieve technique is a special case, where the number of a sub-problems is just .

- 5
- **1 page 34**
- Few

73) Dynamic programming algorithms need to store the results of intermediate sub-problems.

- **True page 75**
- False

74) In counting sort, once we know the ranks, we simply ----- numbers to their final positions in an output array.

- Delete
- **Copy page 57**
- Mark
- Arrange

75) In Quick sort, we don't have control over the sizes of recursive calls.

- **True page 49**
- False
- Less information to decide
- Either true or false

76) Before sweeping a vertical line in a plane sweep approach, in start sorting of the points is done in increasing order of their ----- coordinates.

- **X (page #18)**

- Y
- Z
- X&Y

77) Random access machine or RAM is a/an

- Machine build by Al-Khwarizmi
- Mechanical machine

- **Mathematical model (page #10)**

- Electronics machine

78) The Huffman codes provide a method of encoding data inefficiently when coded using ASCII standard .

- True

- **False(page #99)**

79) A heap is a left-complete binary tree that conforms to the -----.

- Increasing order only
- Decreasing order only

- **Heap order (page# 40)**

- $(\log n)$ order

80) For the sieve technique we solve the problem,

- **Recursively (page #34)**

- Mathematically
- Precisely
- Accurately

81) Median is not useful measure of central tendency of given input set especially when the distribution of values is highly skewed.

- True
- **False (page # 34)**

82) The omega-notation allows us to state only the asymptotic-----bounds.

- Middle
- **Lower (page #25)**
- Upper
- Both middle and upper

83) In plane sweep approach, a vertical line is swept across the 2-D plane and-----structure is used for holding the maximal points lying to the left of the sweep line.

- Array
- Queue
- **Stack (page #18)**

- Tree

84) When a heapify procedure is applied to the root node to restore the heap, then at each level, the comparison performed takes time.

- It will take $O(\log n)$
- It cannot be predicted
- **It will take $O(1)$ (page #43)**
- Time will vary according to the nature of input data

85) Efficient algorithm requires less computational-----.

- Memory
- Running time
- **Memory and running time (page #9)**
- Energy

86) Recurrence can be described in terms of a tree.

- **Yes (page#31)**
- No

87) Time complexity of dynamic programming based algorithm for computing the minimum cost of Chain Matrix multiplication is-----.

- $\log n$
- n
- n^2 (n square)

• **n^3 (n cube) (page#90)**

88) In order to say anything meaningful about our algorithms, it will be important for us to settle on a-----
-----.

- Java programming
- C++ programming
- Pseudo program

• **Mathematical model of computation (page # 10)**

89) The iteration method is used for -----.

- Comparing sorting algorithms only

• **Solving recurrence relation (page # 31)**

- Merging elements in merging sort
- Divide elements in merge sort

90) Chain matrix multiplication problem can be solved through-----strategy.

• **Dynamic programming (page #85)**

- Greedy
- Divide and conquer
- Sorting

91) For comparison-based sorting algorithms, it is -----possible to sort more efficiently than $\Omega(n \log n)$ time.

- Always

• **Not (page #54)**

- Sometimes not
- Sometimes

92) Dynamic Programming approaches are usually useful in solving optimization problems.

- **True (page # 97)**

- False

93) In sorting the key value or attribute-----from an ordered domain.

- **Must be (page #39)**

- Not always

94) Result of asymptotical analysis of $n(n-3)$ and $4n*n$ is that-----.

- $n(n-1)$ is asymptotically less
- $n(n-1)$ is asymptotically Greater
- both are asymptotically Not equivalent

- **both are asymptotically Equivalent (page #23)**

95) Floor and ceiling are-----to calculate while analyzing algorithms.

- Very easy

- **Usually consider difficult pg#31**

96) ----- of reference is an important fact of current processor technology.

- Defining
- Assigning
- Formality

- **Locality (page # 8)**

97) As per algorithm of dynamic programming , we need to store

- First sub- problem only
- Best solution only

- **Intermediate sub-problems (page #75)**

- Final solution only

8) In Sieve technique, we solve the problem-----.

- **In recursive manner (page # 34)**

- Non recursively
- Using Merge sort algorithm
- Using Brute force technique

99) Upper bound requires that there exist positive constant c_2 and n_0 such that $f(n) \leq c_2 n$ for all $n \geq n_0$.

- Less than
- **Equal to or less than (page # 24)**
- Equal to or greater than
- Greater than

100) In 2-D matrix problem, A point p is said to be dominated by point q if-----.

- $P.x = q.x$ and $p.y = q.y$
- $P.x = q.x$ and $p.y < q.y$
- $P.x > q.x$ and $p.y > q.y$
- **$P.x \leq q.x$ and $p.y \leq q.y$ (page # 11)**

101) A Random Access Machine (RAM) is an idealized machine with----- random access.

- **Infinite large (page # 10)**

- 512 MB
- 256 MB
- 2 GBs

102) There are-----entries in the Edit Distance Matrix.

- $\Theta(n)$
- **$\Theta(n^2)$ (page # 84)**
- $\Theta(n+2)$
- $\Theta(n+100)$

103) Counting sort is suitable for sorting the elements within range 1 to P , where-----.

- P is large
- **P is small (page # 57)**
- P is very large
- P is undetermined

104) While analyzing searching algorithm, we make a number of passes, in fact it called----

- $n(n+1)$
- **$\log(n)$ (page #37)**
- $n/3$

- $n/4$

105) It is not a Fibonacci Sequence.

1,1,1,2,3,5,8,13,21,34,

55,----

- **True (page #73)**

- False

106) In Random Access Machine (RAM), instructions are executed in-----.

- Parallel
- Batch

One by one (page # 10)

- Multiple times

107) Dynamic programming formulation of the matrix chain multiplication problem will store the solutions of each sub problem in a(n):

- Array

Table (page #86)

- Variable
- Class

108) An algorithm is a mathematical entity, which is independent of-----.

- Programming language
- Machine and programming language
- Compiler and programming language

Programming language Compiler and Machine (page #7)

109) In Fibonacci sequence, unnecessary repetitions do not exist at all.

True (page# 74)

- False

110) In Quick sort algorithm, pivots form-----.

- Stack

Queue

- Graph

111) In Heap Sort Algorithm, Heapify procedure is----- in nature.

- **Recursive (page #43)**

- Non-recursive

112) In Bucket sort, if there are duplicates then each bin can be replaced by a :

- Stack

Link list (page #69)

- Hash table
- Heap

113) Counting sort assumes that the numbers to be sorted are in the range -----.

- k to n where n is large

1 to k where k is small (page # 57)

- k to n where k is small
- k to n where n is small

114) In Dynamic Programming, our approach is to -----.

- Develop the solution in a top-down fashion.
- Express the problem non-recursively.

o Build the solution in a bottom- up fashion. Pg#92

- Input several sub-problems simultaneously.

115) In knapsack Problem, the goal is to put items in the knapsack such that the value of the items is -----

----- subject to weight limit of knapsack.

- Minimized
- Decreased

Maximized

- None of the given option

116) If there are $\Theta(n^2)$ entries in edit distance matrix then each entry $E(i,j)$ takes ----- time to compute.

- $\Theta(n)$
- $\Theta(n^2)$

$\Theta(1)$ (page #84)

- $\Theta(n \log n)$

117) In ----- knapsack Problem, limitation is that an item can either be put in the bag or not.

Fractional items are not allowed.

- 0
- 1

0/1 (page #91)

- Fractional

118) Heapsort is a/an ----- and ----- sorting algorithms.

- Not in-place, not stable one
- In-place , stable one

In-place, not stable one (page#54)

- Not in-place , stable one

119) Selection sort is not an in-place sorting algorithm.

- True

False (page #54)

120) Memoization is a part of Dynamic Programming strategy.

True (page 74)

- False

127) Quick sort algorithms divide the entire array into-----sub arrays.

- One

Two (Google)

- Three
- Four

121) Insertion sort is an in-place algorithm.

a) True (page #54)

b) False

122) When matrix A of 5x3 is multiply with matrix B of 3x4 then the number of multiplication required is :

- 15
- 12
- 36
- **60** **page 84**

123) If matrix A of dimensions 2x4 is multiplied with matrix B of dimensions 4x3, then the dimensions of resultant matrix is :

- 2x4
- 4x3
- 3x4
- **2x3** **page 84(conceptual)**

124) Dynamic Programming strategy is useful when sub-problems are independent.

- True

False google

125) If Matrix-A of dimension 2x4 multiplied with Matrix-B of dimension 4x3, then the dimension of resultant Matrix is:

- 2x4
- 4x3
- 3x4
- **2x3** **page 84(conceptual)**

126) In the following code the statement "cout<< i" executes-----

----times. **for (int i=1; i<=n; i++)**

cout<< i;

- **n times** **(conceptual)**
- n+5 times
- Infinite times
- Zero times

127) Edit distance algorithm based on-----strategy.

- Greedy
- **Dynamic programming** **page 77**
- Divide and conquer
- Searching

128) If Matrix A has dimensions "3x2" and Matrix B has dimensions "2x3", then multiplication of Matrix-A and Matrix-B will result a new Matrix-C having dimensions.

- 3x2
- 2x3
- 2x2
- **3x3 page 84(conceptual)**

129) The function $f(n) = n(\log n + 1)/2$ is asymptotically equivalent to $n \log n$. Here Lower Bound means function f

(n) grows asymptotically at ----- as fast as $n \log n$.

- **Least page 24**
- Normal
- Most
- All

130) Single atom from a larger set of -----.

- Constant
- Pointers
- Phases
- **n items page 34**

131) Fibonacci sequence was named on a-----, a famous mathematician in 12th century.

- Fred brooks
- Grady booch

Leonardo Pisano (page # 73)

- Edgar F.Codd

132) In dynamic programming, our approach is to-----.

- Develop the solution in a top down fashion
- Express the problem non-recursively

Build the solution in a bottom up fashion (page #92)

- Input several sub-programs simultaneously

133) In heap sort algorithm, we build-----for ascending order.

- **Max heap page 42(conceptual)**
- Min heap

134) Bubble sort is not an in-place sorting algorithm.

- True

False (page #54)

135) In partition algorithm, the subarray-----has elements which are greater than pivot element X.

- $A[p \dots r]$
- $A[p \dots q-1]$
- $A[q]$

$A[q+1 \dots r]$ (page #46)

136) In Heap sort algorithm, if heap property is violated-----.

- We call build heap procedure

We call heapify procedure (page # 41)

- We ignore
- Heap property can never be violated

137) -----is not a characteristics of Random Access Machine.

- Single-processor
- Assigning a value to a variable

Locality of reference (page #10)

- Executing an arithmetic instruction

138) Boolean operation is a-----operation on an idealized RAM model of computation.

- Starting

Basic (page #10)

- Advance
- Normal

139) How much time merge sort takes for an array of numbers?

- $T(n^2)$
- $T(n)$
- $T(\log n)$

- **$T(n \log n)$ page 4**

140) Sorting is one of the-----problems in Computer Sciences.

Easiest page 39

- Solution-less
- Least popular
- Tony Hoare

142) We can make-----recursive calls in Fibonacci sequence.

- **Infinite google**
- Finite

140) Insertion sort is a ----- sorting algorithm.

- Unstable
- **In-place page 54**
- Not in-place
- In-portion

141) We do not need to prove comparison- based sorting algorithms by mathematically , it always takes-----

-----:

- Big Oh $n \log(n)$
- **Omega $n \log(n)$ page 54**
- Omega $n(n^2)$
- Theta $n \log(n)$

142) A Principal operation for maintaining the heap property is called heapify, it is also called :

- **Shifting Down page 43**
- Shifting Up
- Shifting Right
- Shifting Left

143) The number of nodes in a complete binary tree of height h is:

- $2^{(h+1)}-1$
- $2^{(h+1)}-1$
- **$2^{(h+1)}$ page 45**
- $((h+1)^2)-1$

144) After partitioning array in Quick sort, pivot is placed in a position such that:

- Pivot is the first element of array

- Pivot is the last element of array
- **Values smaller than pivot are on left and larger than pivot are on right**
page 35

- Values larger than pivot are on left and smaller than pivot are on right

145) ----- programming is essentially recursion without repetition:

- Fast
- **Dynamic** **page 75**
- Array
- $N \log n$

146) We can use the _____ property to devise a recursive formulation of the edit distance problem.

- small substructure
- algorithmic

real

- **optimal substructure** **page 78**

147) Applying the sieve technique to selection problem, _____ element is picked from array.

- Output
- Total
- Input
- **Pivot** **page 35**

148) If matrix A of dimension $p \times q$ is multiply with matrix B of dimension $q \times r$, then the dimension of the resultant matrix is:

- $p \times q$
- **$p \times r$** **page 84**
- $q \times r$
- $r \times p$

149) In knapsack problem, each item must be entirely accepted or rejected, is called _____ problem.

- Fractional
- **"0-1"** **page 92**
- Linear
- Optimal

150) In the case of _____ analysis does not depend upon on the distribution of input.

- Merge sort
- **Quick sort** **page 50**
- Insertion sort
- Heap sort

- 151) In Heapsort Algorithm, total time taken by heapify procedure is:
- **$O(\log n)$ page 43**
 - $O(\log^2 n)$
 - $O(n \log n)$
 - $O(n^2 \log n)$
- 152) In merge sort algorithm, we split the array _____ to find index q.
- from start
 - **midway page 28**
 - from end
 - both from start or end
- 153) Al-Khwarizmi was a/an _____
- artist
 - **mathematician page 7**
 - astronomer
 - khalifah
- 154) In Sieve Technique, we know the item of interest.
- True
 - **False page 34**
- 155) Radix sort is a _____ integer sorting algorithm.
- Comparative
 - **Non-comparative google**
 - In-Place
 - Unstable
- 156) Random Access Machine (RAM) can execute _____ instruction.
- only logical
 - parallel
 - only arithmetic
 - **logical and arithmetic page 10**
- 157) Which of the following is calculated with Big Omega notation?
- Medium bounds
 - **Upper bounds page 25**
 - Lower bounds
 - Both upper and lower bounds

158) Asymptotic growth of $8n^2 + 2n - 3$ is:

- $O(n^2+n)$
- **$O(n^2)$ page 23**
- $O(Bn^2)$
- $O(Bn^2+2n)$

159) _____ is the process of avoiding unnecessary repetitions by writing down the results of recursive calls and looking them up again if we need them later.

- Loop
- **Memoization page 74**
- Recursion
- Function

160) In Heap Sort algorithm, the first step is to _____.

- **Call Build-Heap procedure page 43**
- Sort the array in descending order
- Call Heapify procedure
- Find the number of input elements

161) The time assumed for each basic operation to execute on RAM model of computation is _____.

- Infinite
- Continuous
- **Constant page 10**
- Variable

162) If input "n" is odd, then median will be _____.

- **$(n+1)/2$ page 34**

- $n/2$
- $(n-1)/2$
- $n+2$

163) In quick sort algorithm, _____ decides nature of Binary Search Tree formed by pivots.

- middle element from input
- smallest element from input
- **rank of the pivot page 49**
- largest element from input

164) The brute-force algorithm for 2D-Maxima runs in order $O(\text{-----})$ time:

- n
- $n(\log n)$

$n*n$

page 18

- n^3

165) In average-case time analysis of quick sort algorithm, the most balanced case for partition is when we divide

the list of elements into-----

- Equal no. of pieces as input elements
- Single piece exactly
- **Two nearly equal pieces google**
- Three nearly equal pieces

166) The main purpose of mathematical analysis is measuring the ----- required by the algorithm:

- Space
- **Execution time page 13**
- Inputs & outputs
- Execution time and memory

167) Dynamic Programming algorithms often use some kind of ----- to store the results of intermediate Sub-problems:

- **Table page 75**
- Variable
- Stack
- Loop

168) The average case running time of quick sort algorithm is theta -----:

- $\log(n)$
- (n)
- **$n\log(n)$ page 53**
- $n \times n \log(n)$

169) What is the best case time complexity of merge sort?

- $O((n^2 \log n))$
- $O((n \log^2))$
- **$O((n \log n))$ page 40**
- $O((n^2))$

170) While solving Selection problem, in Sieve technique we partition input data -----:

- In increasing order
- In decreasing order
- **According to Pivot page 35**
- Randomly

171) In average-case time the probability of seeing input is denoted by-----:

- $P\{I\}$
- $P[I]$

- **P(I) page 13**
- $P<I>$

172) To check whether a function grows faster or slower than the other function, we use some asymptotic notations, which is -----:

- Big-oh notation
- Theta notation
- Omega notation
- **All of the given google**

173) Asymptotic notations are used to describe ----- of an algorithm:

- Length
- **Running time (google)**
- Size
- Compile time

174) In recursive formulation of Knapsack Problem

$V[0,j] = \text{-----}$ for $j \geq 0$

- -1
- **0 (page #93)**
- 1
- 2

175) If a matrix has three rows and two columns, then dimensions of matrix will be:

- **3x2 (google0)**
- 2x2
- 2x3
- 3x3

176) Selection sort takes theta --- in the worst case:

(n²)

- (n)
- Log(n)
- nlog(n)

177) Which of the following problems can be solved using dynamic problem?

- Bubble sort problem
- **Matrix chain multiplication problem page 92**
- Greedy search problem
- Fractional knapsack problem

178) ----- algorithm based on Dynamic Programming strategy:

- Heap sort
- Binary tree
- Quick sort
- **Edit distance page 77**

179) The definition of theta-notation relies on proving ----- asymptotic bound:

- One
- Lower
- Upper
- **Both lower and upper page 25**

180) In order to say anything meaningful about our algorithms, it will be important for us to settle on a -----:

- Java program
- C++ program
- Pseudo program
- **Mathematical model of computation page 10**

181) In asymptotical analysis of $n(n-3)$ and $4n*n$, as n becomes large, the dominant (fastest growing) term is some Constant times-----,

- $n+1$
- $n-1$
- n
- **$n*n$ page 23**

182) ----- is in- place sorting algorithm

- **Bubble sort page 54**
- Merge sort
- Linear sort
- Binary sort

183) In chain matrix multiplication, if there are ----- ways in which outer most pair of parentheses can placed:

- n^2
- $2n$
- $n+1$
- **$n-1$ page 85**

184) In chain matrix multiplication, table is filled ----- to find the multiplication of matrix:

- row wise
- column wise
- **diagonally page 86**

- bottom-to-up

185) Which one sorting algorithm is best suited to sort an array of 2 million elements?

- Bubble sort
- Insertion sort
- Merge sort
- **Quick sort google**

186) The problem with the brute-force algorithm is that it uses ----- in pruning out decisions:

- Worst-case time
- **No intelligence page 18**

- Outside looping
- Artificial intelligence

187) Quicksort is a/an ----- and ----- sorting algorithm:

- No in-place , not-stable one
In-place , not stable one page 54

- In-place , stable one
- Not in-place , stable one

191) In Heap Sort algorithm, the maximum levels an element can move upward is -----:

- **Theta (log n) page 43**

- Big-oh(log n)
- Omega (log n)
- $O(1)$ i.e. Constant time

192) If the time complexity of an algorithm is $O(n)$, then it is called ----- time complexity:

- **Linear page 37**
- Constant
- Average
- Exponential

193) Dynamic Programming approach is usually useful in solving ----- problems:

- Normal
- **Optimization page 97**
- Array
- Loop

194) What time does Merge Sort algorithm take in order to sort an array of 'n' numbers?

- $\Theta(n)$
- $\Theta(\log n)$
- $\Theta(n^2)$
- **$\Theta(n \log n)$ page 30**

CS-502 (Fundamentals of Algorithms)

Grand Quiz - Fall 2020

If you found any error in the file, Please correct it accordingly.

163) The sequence of merge sort algorithm is:

- Divide Combine-Conquer
- Conquer-Divide-Combine
- c. **Divide-Conquer-Combine Page 27**
- Combine-Divide-Conquer
- In _____ Knapsack Problem, limitation is that an item can either be put in the bag or not. Fractional items are not allowed.

0

1

c. **0/1 Page 91**

Fractional

165) In Selection algorithm, we assume pivot selection takes theta _____ running time.

a. n Page - 36

n^2

n^3

$\log(n)$

- In Heap Sort algorithm (using max heap), when every time maximum elements removed from top _____.

We call merge Sort Algorithm

it becomes Order n^2 Algorithm

Divide and Conquer strategy helps us

d. We are left with a hole Page – 41

- 167) If matrix A of dimension $p \times q$ is multiply with matrix B of dimension $q \times r$, then each entry in resultant matrix takes _____ time.

a. $O(q)$ Page - 84

168) $O(1)$

169) $O(p \times q)$

170) $O(q \times r)$

169) _____ is a method of solving a problem in which we check all possible solutions to the problem to find the solution we need.

- Plane-Sweep Algorithm
- Sorting Algorithm

c. **Brute-Force Algorithm** **google**

- Greedy approach

171) The worst case running time of Quick sort algorithm _____.

- Cannot be quadratic

b. **Is quadratic** **google** [QuickSort and its Analysis \(codesdope.com\)](https://www.codesdope.com/blog/view/quick-sort-algorithm-analysis) &

<http://personal.kent.edu/~rmuhamma/Algorithms/MyAlgorithms/Sorting/quickSort.htm>

Is always Exponential

Is linear

172) In max heap (for Heap Sort algorithm), when every time maximum element is removed from top we replace it with _____ leaf in the tree.

- second last

b. **Last** **Page -41**

- First
- Any

174) Quick sort algorithm was developed by -

AlferdAho

Sedgewick

John Vincent Atanasoff

d. **Tony Hoare** **– Google wikipedia**

- If Matrix-A has dimensions "3x2" and Matrix-B has dimensions "2x3", then multiplication of Matrix-A and Matrix-B will result a new Matrix-C having dimensions.

3x2

2x3

2x2

d. 3x3

<http://www.calcul.com/show/calculator/matrix-multiplication>

- For comparison-based sorting algorithms, it is possible to sort more efficiently than Omega $n \log(n)$ time.
 - a. Always
 - b. **Not** **P-54**
 - c. Sometimes

Sometimes not

176) Dynamic Programming approach is usually useful in solving optimization problems.

a. True

https://en.wikipedia.org/wiki/Dynamic_programming &

<http://web.mit.edu/15.053/www/AMP-Chapter-11.pdf>

178) False

179) In Sorting the key value or attribute_____ from an ordered domain.

a. Must be page 39

- Not always
- May be
- Occasionally

181) Result of asymptotical analysis of $n(n-3)$ and $4n*n$ is that _____

- $n(n-1)$ is asymptotically Less
- $n(n-1)$ is asymptotically Greater
- Both are asymptotically Not equivalent

d. Both are asymptotically Equivalent page 23 ($4n*n = 4n^2$)

182) Floor and ceiling are _____ to calculate while analyzing algorithms a. Very easy

b. Usually considered difficult

P-31

3rd Option is missing

4th Option is missing

184) _____ of reference is an important fact of current processor technology.

- Defining
- Assigning

- Formality

d. Locality

P-8

185) In max-heap, largest element is stored at root node. Where is the smallest element stored?

- Right Node

Leaf Node

- Middle Node
- Left Node

- In average-case time analysis of Quick sort algorithm, the most balanced case for partition is when we divide the list of elements into _.

Equal no. of pieces as of input elements

Single piece exactly

Two nearly equal pieces

Three nearly equal pieces

- Which of the following is calculated with Big O notation?

Medium bounds

b. **Upper bounds**

Page - 25

- Lower bounds
- Both upper and lower bounds

187) Edit distance algorithm based on _____ strategy

- Greedy

b. **Dynamic Programming**

Page - 81

Divide and Conquer

Searching

- In Heapsort Algorithm, total time taken by heapify procedure is _____

a. $O(\log n)$

Page-43

- $O(\log^2 n)$
- $O(n \log n)$
- $O(n^2 \log n)$

193) Al-Khwarizmi was a/an _____

- Artist

b. Mathematician

P-7

Astronomer

Khalifah

194) When matrix A of 5×3 is multiplied with matrix B of 3×4 then the number of multiplications required is: Not found exactly

15

12

36

60 Not Found exactly but as per formula at page 84,

24. Pseudo code of algorithms are to be read by _____.

a. **People** **Page -12**

RAM

Computer

Compiler

195) The sieve technique is a special case, where the number of sub-problems is Just

a. **1** **P-34**

b. 2

c. 3

d. 4

26. When a recursive algorithm revisits the same problem over and over again, we say that the optimization problem has _____ sub-problems.

a. **Overlapping** **– Google Search**

b. Over costing

c. Optimized

d. Three

27. Sieve technique is very important special case of Divide-and-Conquer strategy.

a. **True** **P-34**

b. False

28. In order to say anything meaningful about our algorithms, it will be important for us to settle on a _____.

a. Java Program

- b. C++ Program
- c. Pseudo program

d. Mathematical model of computation

P-
1
0

29. Merge sort is based on _____.

- a. Brute-force
- b. Plan-sweep
- c. Axis-sweep

d. Divide and Conquer

P-
27

30. What time does Merge Sort algorithm take in order to sort an array of 'n' numbers?

- a. (n)
- b. (log n)
- c. (n²)

d. (n log n) [Google Search](#)

31. In Heap Sort algorithm, the first step is to _____.

a. Call Build-Heap

[a. procedure Page - 46](#)

- b. Sort the array in descending order
- c. Call Heapify procedure
- d. Find the number of input elements

32. The definition of theta-notation relies on proving _____ asymptotic bound.

- a. One
- b. Lower
- c. Upper

[b. d. Both lower & upper Page - 25](#)

33. In merge sort algorithm, to merge two lists of size n/2 to a list of size n, takes _____ time.

a. [Theta \(n\) Page - 32](#)

- b. Theta log(n)
- c. Theta log₂(n)
- d. Theta n log(n)

34. We can make _____ recursive calls in Fibonacci Sequence.

- a. Infinite
- b. **Finite** google
- c. Only one
- d. Zero

35. Following is NOT the application of Edit Distance problem.

- a. Speech recognition
- b. Spelling Correction
- c. **Ascending Sort** Page - 76
- d. Computational Molecular Biology

36. In plane sweep approach, a vertical line is swept across the 2d-plane and structure is used for holding the maximal points lying to the left of the sweep line.

- a. Array
- b. Queue
- c. **Stack**
- d. Tree

Page - 18

37. When a heapify procedure is applied to the root node to restore the heap, then at each level, the comparison performed takes time:

- a. It will take $(\log n)$.
- b. It can not be predicted

c. **It will take $O(1)$.**

Page - 43

d. Time will vary according to the nature of input data.

38. _____ time is the maximum running time over all legal inputs.

a. **Worst-case**

Page - 13

- b. Average-case
- c. Best-case
- d. Good-case

39. Efficient algorithm requires less computational...

- a. Memory
- b. Running Time

c. **Memory and Running Time**

Page - 9

d. Energy

40. For average-case time analysis of Quick sort algorithm, Pivot selection is on average basis from _____

a. half of the input values

b. all possible random values Page - 50

c. Pivot is input separately

d. values greater than 5

41. Selection algorithm takes theta _____

a. (n^2)

b. (n) Page - 37

c. $\log(n)$

d. $n \log(n)$

42. Recurrence can be described in terms of a tree.

a. **Yes** **Page - 31**

b. No

43. Time complexity of Dynamic Programming based algorithm for computing the minimum cost of Chain Matrix Multiplication is _____

a. Log n

b. n

c. n^2 (n square)

d. **n^3 (n cube)** **Page -90**

44. The Iteration method is used for _____

a. Comparing sorting algorithms only

b. **Solving Recurrence relations** **Page 31**

c. Merging elements in Merge sort

d. Dividing elements in Merge sort

45. In 3-Dimensional space, a point P has _____ coordinate(s).

a. (X, Y)

b. (X, 0)

c. (0, Y)

d. **(X, Y, Z)**

46. Chain matrix multiplication problem can be solved through _____ strategy.

a. **Dynamic programming** **Page - 85**

- b. Greedy
- c. Divide and conquer
- d. Sorting

47. Merge sort have running time....running time of Heap sort. Not found exactly

- a. Greater than
- b. Less than
- c. Equal to
- d. Different than

Google

48. Median is not useful measure of central tendency of given input set especially when the distribution of values is highly skewed.

a. True

b. False

Page – 34

49. We do not need to mathematically prove that for comparison-based sorting algorithms always takes $\Omega(n \log n)$ time.

a. True

Google & VU Tech

(pg 46 not very clear)

b. False

50. The Omega-notation allows us to state only the asymptotic _____ bounds.

a. Middle

b. Lower

Page 25

c. Upper

d. Both lower & upper

51. Both lower & upper Sorting can be in _____

a. Increasing order only

b. Decreasing order only

Both Increasing and Decreasing order – (My opinion)

c. Random order

52. Radix sort performs sorting the numbers _____ digit (s) at a time.

a. **One** **Page - 71**

b. Two

c. Three

d. All

53. Quicksort is a/an _____ and _____ sorting algorithm.

a. Not in place, not stable one

b. **In place , not stable one** **Page - 54**

c. In place , stable one

d. Not in place , stable one

54. Consider three matrices X,Y,Z of dimensions 1x2, 2x3,3x4 respectively. The number of multiplications of (XY) Z is:

a. **18** **As per lecture slides**

b. 32

c. 24

d. 30

55. In Fibonacci Sequence, unnecessary repetitions do not exist at all.

a. True

b. **False** **Page – 74**

56. It is not a Fibonacci sequence . 1,1,1,2,3,5,8,13,21,34,55,.....

a. True Page - 73

b. False

57. Heap sort is a/ an _____ and _____ sorting algorithm.

a. Not in place, not stable one

b. In place , not stable one Page - 54

c. In place , stable one

d. Not in place , stable one

58. Identify the True Statement

a. The knapsack problem does not belong to the domain of optimization problems.

b. The knapsack problem belongs to the domain of optimization problems. Page - 91

c. The Knapsack problem cannot be solved by using dynamic programming

d. The knapsack problem is optimally solved by using brute force algorithm.

59. In Dynamic Programming, our approach is to _____

a. Develop the solution in a top-down fashion

b. Express the problem non-recursively

c. Build the solution in a bottom-up fashion Page - 75

d. Input several sub-problems simultaneously

60. Counting sort is suitable to sort the elements in range 1 to K;

a. K is large

b. **K is small** Page - 57

c. K may be large or small

d. None

61. We can multiply two matrices A and B only when they are compatible which means

a. **Number of columns in A must be equal to number of rows in B.**

it seems Correct as per page 84

(https://www.varsitytutors.com/hotmath/hotmath_help/topics/matrix-multiplication#:~:text=You%20can%20only%20multiply%20two,an%20m%C3%97p%20matrix.)

b. Number of rows and columns do not matter

c. Number of columns in A must be equal to number of columns in B

d. Number of rows in A must be equal to number of rows in B

62. Matrix multiplication is a (n) _____ operation.

a. Commutative

b.

c. **b. Associative Page 85**

c. Neither commutative nor associative

d. Commutative but not associative

63. In Dynamic Programming approach, solution is modified / changed

a. Always once

b. **At each stage** google and wikipedia

c. Only for specific problems

d. At 4th stage only

64. In Knapsack problem, the goal is to put items in the Knapsack such that the value of the items is _____ subject to weight limit of knapsack.

a. Minimized

b. Decreased

c. **Maximized** Page - 91

d. None of the given options

65. An in-place sorting algorithm is one that _____ uses additional array for storage.

a. Always

b. Permanently

c. **Does not** Page - 54

d. Sometime

66. Memoization is a part of Dynamic Programming Strategy.

a. **True** Page - 74

b. False

67. If matrix A of dimension 2x4 is multiply with matrix B of dimension 4x3, then the dimension of resultant matrix is Not found exactly

a. 2x4

b. 4x3

c. 3x4

d. **2x3** It seems correct as per second last Para of page 84

68. In Dynamic Programming approach, we do not store the solution to each sub-problem in case if it reappears.

a. True

b. False Page - 75

69. Dynamic Programming is a problem-solving approach in which____

a. Problem is solved in Zero time

b. Solution is developed only at final stage

c. Both are correct

d. Both are incorrect google

70. In Fibonacci sequence, each term is calculated by____ previous__ terms.

a. Subtracting, Two

b. Adding, Three

c. Adding, Two Page - 73

d. Multiplying, Two

71. Selection sort is not an in-place sorting algorithm.

a. True Page - 54

b. False

72. If there are $\theta(n^2)$ entries in edit distance matrix then the total running time is:

a. $\theta(n)$

b. $\theta(1)$

c. $\theta(n^2)$ Page – 84

d. $\theta(n \log n)$

73. The only way to convert a string of i characters into the empty string is with i deletions, represented as

a. $E(0,j) = j$

b. $E(i,j) = 1$

c. $E(0,i) = j$

d. $E(i,0) = i$ Page - 78

74. Dynamic programming formulation of the matrix chain multiplication problem will store the solutions of each sub problem in an

a. Array

b. Table Page - 86

c. Variable

d. class

75. We can use the optimal substructure property to devise a formulation of the edit distance problem.

- a. Selective
- b. Optimum
- c. Iterative
- d. **Recursive**

Page - 78

76. Sorting is performed on the basis of _____.

- a. Computational resources
- b. Asymptotic notation
- c. Summation

d. Some key value of attribute

page- 39

77. In Heap Sort algorithm, we call Build-heap procedure _____.

- a. **Only once**
- b. Twice
- c. Thrice
- d. As many times as we need

page 46

78. Radix sort is not a non-comparative integer sorting algorithm.

- a. **True**
- b. False

Google Search

79. In the statement “output P[1].x, P[1].y”, the number of times elements of P are accessed is _____.

a. 1

b. 2

page 14

c. 3

d. 4

80. The main purpose of mathematical analysis is measuring the _____ required by the algorithm.

a. Space

b. Execution time

P-13

c. Inputs & outputs

d. Execution time and memory

81. _____ provides us more accurate result when input values are not closer with each other

a. Average

Median (p-34)

c. Mode

d. Mean

82. The process of _____ ends when you are left with such tiny pieces remaining that it is trivial to solve them.

a. Brute-force

b. Plan-sweep

c. Divide and Conquer

P-27

d. Axis-sweep

83. _____ overcomes the limitations of _____ by working as per positional notations of numbers.

a. Counting sort, Radix sort

b. Radix sort, Counting sort

P-71

84. Memorization is a part of Dynamic Programming strategy.

a. True

P-74

b. False

85. Rank of an element can be defined as _____.

a. One minus the number of elements that are smaller

b. Two plus the number of elements that are greater

One plus the number of elements that are smaller P-34

c. Two minus the number of elements that are smaller

86. If the time complexity of an algorithm is given by $O(1)$, then its time complexity would be

a. Polynomial

b. Exponential

c. Constant - Wikipedia

d. Average

87. Quick sort is a recursive algorithm.

a. True Wikipedia ; Google

b. False

88. The asymptotic growth of $n(n+1)/2$ is:

$O(n^2)$ As the n^2 term has the largest contribution, the Big-O complexity is $O(n^2)$

a. $O(n)$

b. $O(n+2)$

c. $O(n \log n)$

89. Approach of solving geometric problems by sweeping a line across the plane is called _____ sweep.

a. Line

b. Plane Page 18

c. Cube

d. Box

90. As per algorithm of Dynamic Programming, we need to store

a. First sub-problem only

b. Best solution only

c. Intermediate sub-problems Pg:75

d. Final solution only

91. In Sieve technique, we solve the problem

a. In recursive manner Pg:34

b. Non recursively

c. Using Merge Sort algorithm

d. Using Brute force technique

92. One of the limitation in 0/1 knapsack is that an item can either be _____ in the bag or not.

a. Use

b. Put Pg:91

c. Move

d. Store

93. Which one is not passed as parameter in Quick sort algorithm?

a. End of the array

b. Middle of the array

c. Array (containing input elements) Google

d. Start of the array

94. In the analysis of Selection algorithm, we get the convergent _____

a. Harmonic

b. Linear

c. Arithmetic

d. Geometric Pg:37

95. A Random Access Machine (RAM) is an idealized machine with random access memory.

a. Infinite large Pg:10

b. 512 MB

c. 256 MB

d. 2 GBs

96. While analyzing Selection algorithm, we make a number of passes, in fact it could be as many as

a. $n(n+1)$

b. $\log(n)$ Pg:37

c. $n/3$

d. $n/4$

97. In Random Access Machine (RAM), instructions are executed in

a. Parallel

b. Batch

c. One by One

Pg:10

d. Multiple times

98. In selection problem, the rank of an element will be its _____ position

a. First

b. final

Pg:34

c. Second last

d. Last

99. The worst-case running time of Merge sort is _____ in order to sort an array of n elements.

a. $O(\log n)$

b. $O(n)$

c. $O(n \log n)$

page 40 and google

d. $O(n)$

100. $f(n)$ and $g(n)$ are asymptotically equivalent. This means that they have essentially the same _____.

a. Results

b. Variables

c. Size

d. Growth rates

P:23

101. An algorithm is a mathematical entity. Which is independent of _____.

a. Programming language

b. Machine and Programming language

c. Compiler and Programming language

d. Programming language Compiler and Machine

P:
07

102. In Quick sort algorithm, Pivots form ____

- a. Stack
- b. Queue

c. Binary Search Tree

P:49

d. Graph

103. Counting sort is suitable for sorting the elements within range 1 to P. where

a. P is large

b. P is small

P-57

c. P is very large

d. P is undetermined

104. In asymptotical analysis of $n^{(5/2)-3}$, as n becomes large, the dominant (fastest growing) term is some constant times

- a. n_1
- b. n
- c. $n+1$

d. $n*n$

P-23

105. ____ Items are not allowed in the 0/1 knapsack. a. Lighter

b. Fractional

P-91

c. Whole

d. Weighty

106. Fibonacci Sequence was named on _____, a famous mathematician in 12th Century.

- a. Fred Brooks
- b. Grady Booch

c. Leonardo Pisano

P-73

- d. Edgar F. Codd

107. In Heap Sort algorithm, we build _____ for ascending sort.

a. Max heap

P-41

- b. Min heap

108. Bubble sort is not an in-place sorting algorithm.

- a. True

b. False

P-54

109. In partition algorithm, the subarray _____ has elements which are greater than pivot element x.

- a. $A[p \dots r]$
- b. $A[p \dots q-1]$
- c. $A[q]$

d. $A[q+1 \dots r]$

P-46

110. In Heap Sort algorithm, if heap property is violated

a. We call Build heap procedure

P-43

- b. We call Heapify procedure
- c. We ignore

d. Heap property can never be violated

111. _____ is not a characteristic of Random Access Machine.

a. Single-Processor

P-10

b. Assigning a value to a variable

c. Locality of reference

d. Executing an arithmetic instruction

112. The only way to convert an empty string into a string of j characters is by doing j insertions, represented as _____

a. $E(i,j) = 1$

b. $E(l,0) = l$

c. $E(0,j) = j$

page 78

d. $E(1,j) = j$

113. In Selection problem, the Sieve technique works in _____.

- a. Non-recursive manner
- b. Constant time
- c. **Phases** **page 34**
- d. One complete go

114. Algorithm is a sequence of computational steps that ---- the input into output.

- a. Merge
- b. Assign
- c. **Transform** **page 7**
- d. Integrate

115. If p_j dominates p_i and p_i dominates p_h then p_j also dominates p_h , it means dominance relation is

- a. **Transitive** **page 18**
- b. Non Transitive
- c. Equation
- d. Symbolic

116. **To find maximal** points in brute-force algorithm each point of the space is compared against _____ of that space.

- a. One other point
- b. **All other points** **page 11**
- c. Few other points
- d. Most of the other points

117. In the following code the statement "cout<<j;"executes ----
----- times. for (j=1; j<=5; j = j+2)

cout<<j;

a. 5 times

b. 2 times

c. 3 times

d. 0 times

118. In merge sort algorithm, we split the array around the _____ index q. a. Entering

e. b. Mid page 17

c. Exiting

d. Summing

119. In Selection problem, the Sieve technique _____.

a. Add some more input items each time

b. Do not work recursively

c. Do not uses Divide and Conquer approach

f. Eliminates undesired data items each time

120. Consider three matrices X, Y, Z of dimensions 1×2 , 2×3 , 3×4 respectively. The number of multiplications of $X(YZ)$ is .

a. 16

b. 32

c. 26

g. d. 32 page 84

121. In Heap Sort algorithm, the total running time for Heapify procedure is

a. Theta ($\log n$)

b. Order ($\log n$)

c. Omega ($\log n$)

h. $O(1)$ i.e. Constant time

122. The sieve technique works where we have to find _____ items(s) from a large input.

a. Single page 34

Two

b. Three

c. Similar

123. In Dynamic Programming based solution of Knapsack Problem, if we decide to take an object i , then we gain_____

a. W (Total Weight of Knapsack)

b. V (Total Value of all items)

c. v_i (Value of object i) page 93

d. None of the given option

124. While Sorting, the order domain means for any two input elements x and y _____ satisfies only.

a. $x < y$ page 39

b. $x > y$

c. $x = y$

d. All of the above

125. For solving Selection problem, we introduced Sieve technique due to _____

a. Using Decrease and Conquer strategy page 34

b. Avoiding to sort all input data

c. Eliminating Rank of an element

d. Using Brute-force approach

126. _____ is one of the few problems, where provable lower bounds exist on how fast we can sort.

a. Searching

b. **Sorting** page 38

c. Both Searching & sorting

d. Growing

127. In plane sweep approach, a vertical line is swept across the 2d-plane from_____.

a. Right to Left

b. **Left to Right** page 18

c. Top to Bottom

d. Bottom to top

128. **In generating** Fibonacci sequence, we can avoid unnecessary repetitions by _____ process.

a. Tokenization

b. **Memorization** page 43

c. Randomization

d. Memorization

129. For _____ values of n, any algorithm is fast enough.

a. **Small** page 14

b. Medium

c. Large

d. Infinity

130. Dynamic programming comprises of _____.

- a. Recursion only
- b. Repetition only
- c. Recursion with Repetition

d. No Repetition but Recursion page 75

131. The function $f(n) = n(\log n + 1)/2$ is asymptotically equal to $n \log n$: Here Lower Bound means function $f(n)$ grows asymptotically at ___ as fast as $n \log n$.

a. Least page 23

- b. Normal
- c. Most
- d. At

132. Counting sort has time complexity.

a. $O(n+k)$

b. $O(n)$ page 58

- c. $O(k)$
- d. $O(n \log n)$

133. Due to left complete nature of binary tree, the heap can be stored in

a. Array page 40

- b. Structures
- c. Link List
- d. Stack

134. Single item from a larger set of _____.

a. Constant

b. Pointers

c. Phases

d. n items page 34

135. In the clique cover problem, for two vertices to be in the same group, they must be _____ each other.

a. Apart from

b. Far from

c. Near to

d. Adjacent to page 76

136. How much time merge sort takes for an array of numbers?

- a. $T(n^2)$
- b. $T(n)$
- c. $T(\log n)$

d. $T(n \log n)$ page 40

137. In in-place sorting algorithm is one that uses arrays for storage.

a. No additional array page 54

- b. An additional array
- c. Both of above may be true according to algorithm
- d. More than 3 arrays of one dimension

138. Brute-force algorithm for 2D-Maxima is operated by comparing _____ pairs of points.

- a. Two
- b. Some
- c. Most

d. All page 18

139. While Sorting, the ordered domain means for any two input elements x and y _____ satisfies only.

- a. $x > y$
- b. $x < y$
- c. $x = y$

d. All of the above page 38

140. Quick sort is.

a. Stable & in place

b. Not stable but in place page 54

c. Stable but not in place

d. Some time stable & some times in place

141. Which may be a stable sort?

a. Merger

b. Insertion

c. Both above page 54

d. None of the above

142. For the Sieve Technique we take time.

a. $T(nk)$ page 34

b. $IT(n / 3)$

c. n^2

d. $n/3$

143. Continuation sort is suitable to sort the elements in range 1 to k.

a. K is Large

b. K is not known

c. K may be small or large

d. K is small page 54

144. Asymptotic growth rate of the function is taken over _____ case running time. .

a. Best

b. Worst page 14

c. Average

d. Normal

145. The sieve technique is a special case, where the number of sub problems is just.

a. 5

b. Many

c. 1 page 34

d. Few

146. In Quick sort, we don't have the control over the sizes of recursive calls.

a. True **page 49**

b. False

c. Less information to decide

d. Either true or false

147. Before sweeping a vertical line in plane sweep approach, in start sorting of the points is done in increasing order of their _____ coordinates. .

a. X **page 18**

b. Y

c. Z

d. X , Y

148. Random access machine or RAM is a/an.
- a. Machine build by Al-Khwarizmi
 - b. Mechanical machine
 - c. **Mathematical model** **page 10**
 - d. Electronics machine
149. The Huffman codes provide a method of encoding data inefficiently when coded using ASCII standard.
- a. True
 - b. **False** **page 99**
150. A heap is a left-complete binary tree that confirms to the _____.
- a. increasing order only
 - b. decreasing order only
 - c. **heap order** **page 40**
 - d. log n order
151. If we associate (x, y) integers pair to cars where x is the speed of the car and y is the negation of the price. High y value for a car means a _____ car.
- a. Fast
 - b. Slow
 - c. Expensive
 - d. **Cheap**
152. Which one of the following sorting algorithms is the fastest?
- a. Merge sort
 - b. **Quick sort**
 - c. Insertion sort
 - d. Heap sort

153. Quick sort algorithm divide the entire array into _____ sub arrays.

2

- a. 3
- b. 4
- c. 5

154. In brute force algorithm, we measure running time $T(n)$ based on _____.

- a. Average-case time and best-case time
- b. **Worst-case time and average-case time** page 46
- c. Worst-case time and best-case time
- d. Best-case time and staring-case time

155. For 2D Maxima problem. Plane Sweep algorithm first of all _____.

a. Sorts all points

b. Delete some points

c. Output the elements

d. Pushes all points on stack

156. There are _____ entries in the Edit Distance Matrix

a. $\Theta(n)$

b. $\Theta(n^2)$ page 84

c. $\Theta(n+2)$

d. $\Theta(n + 100)$

157. Which symbol is used for Omega notation?

a. (O)

b. (Θ)

c. (Ω)

c. (@)

158. Selection sort is a _____ sorting algorithm

d. a. In-place page 54

b. Not In-Place

c. Stable

d. in-partition

159. In Dynamic Programming based solution of knapsack problem, to compute entries of 'V', we will imply a(n) _____ approach.

a. Subjective

b. Inductive

- c. Brute Force
- d. Combination

160. We do not need to prove comparison-based sorting algorithms by mathematically. It always takes _____ time.

- a. Big Oh $n \log(n)$
- b. Omega $n \log(n)$
- c. Omega $n(n^2)$
- d. Theta $n \log(n)$

161. Merge sort is a/an _____ and _____ sorting algorithm

- a. Not in-place, not stable one
- b. In-place, not stable one
- c. In-place, stable one

d. Not in-place, stable one page 54

162. Cubic function will _____ a quadratic function.

- a. Prove
- b. be equal to
- c. overtake Page 25
- d. find

163. Insertion sort is a _____ sorting algorithm

- a. Unstable
- b. In-place Page 54
- c. Not In-Place
- d. in-partition

164. To check whether a function grows faster or slower than the other function, we use some asymptotic notations, which is _____.
- a. Big-oh notation
 - b. Theta notation
 - c. Omega notation
 - d. All of the given
165. Asymptotic growth of $8n^2 + 2n - 3$ is:
- a. $\Theta(n^2 + n)$
 - b. $\Theta(n^2)$ page 14
 - c. $\Theta(8n^2)$
 - d. $\Theta(8n^2 + 2n)$
166. In the analysis of algorithms, _____ plays an important role.
- a. text analysis
 - time
 - b. growth rate
 - c. money

167. In inductive approach of knapsack problem, we consider 2 cases, _____

Or _____.

a. Median, Mode

b. Recursive, Iterative

c. Leave object, Take object page 93

d. Sequentially. Parallel

168. Random Access Machine (RAM) can execute _____ instructions

a. only logical

b. parallel

c. only arithmetic

d. logical and arithmetic

169. Using _____ algorithm, efficiency is not given much importance

a. Greedy

b. Merge sort

c. Processing as there is no algorithm by this name

d. Brute Force

170. Bubble sort takes theta _____ in the worst case

a. (n^2) page 39

b. (n)

c. $\log(n)$

d. $n\log(n)$

171. If matrix A of dimension $p \times q$ is multiply with matrix B of dimension $q \times r$, then dimension of resultant matrix is:

a. $p \times q$

b. $p \times r$ page 84

c. $q \times r$

d. $r \times p$

172. Dynamic Programming algorithms often use some kind of _____ to store the results of intermediate sub-problems

a. table (Page 75)

b. variable

c. stack

d. loop

173. _____ is in-place sorting algorithm.

a. Bubble sort (Page 54)

b. Merge sort

c. Linear search

d. Binary Search

174. Which one of the following problems can be solved using dynamic problem?

a. Bubble sort problem

b. Matrix chain multiplication problem page 85

c. Greedy search problem

d. Fractional knapsack problem

175. In chain matrix multiplication, solutions of the sub-problems are stored in a

_____.

a. Array

b. Table page 86

c. Tree

d. Link list

176. What is the average running time of a quick sort algorithm?

a. $O(n^2)$

b. $O(n)$

c. $O(n \log n)$ (Page 49)

d. $O(\log n)$

177. Sorting Algorithms having O _____ running time are considered to be slow ones.

a. (n)

b. (n^2) (Page 39)

c. $(n\log(n))$

d. $(\log(n))$

178. While solving Selection problem, in Sieve technique we partition input data

a. In increasing order

b. In decreasing order

c. According to Pivot

d. Randomly

179. _____ is the process of avoiding unnecessary repetitions by writing down the results of recursive calls and looking them up again if we need them later.

a. Loop

b. Memoization page 74

c. Recursion

d. Function

180. In average-case time the probability of seeing input is denoted by _____.

a. $p\{I\}$

b. $p[I]$

c. $p\langle i \rangle$

d. $p(i)$ page 13

181. While applying the Sieve technique to selection sort, how to choose a pivot element.

a. Through mean

b. Linear

c. Randomly page 35

d. Sequentially

182. Number of _____ of the pseudo code are counted to measure the running time.

a. Inputs

b. Outputs

c. Steps page 13

d. Pages

183. Developing a dynamic programming algorithm generally involves _____ separate steps.

a. One

b. Two page 75

c. Three

d. Four

184. $8n^2 + 2n + 3$ will exceed $c28(n)$, no matter how large we make _____.

a. n

b. $2n$

c. $c2$ page 25

d. this quadratic equation

185. The running time of quick sort algorithm_____.
- a. Is impossible to compute
 - b. Has nothing to do with pivot selection
 - c. Is Random upon each execution
 - d. **Greatly influenced by the selection of pivot** **page 49**
186. _____ involves breaking up the problem into sub problems whose solutions can be combined to solve the global problem.
- a. Complexity Theory
 - b. Dynamic programming solution
 - c. **Divide and Conquer Strategy** **page 34**
 - d. Greedy Algorithms
187. In _____ we have to find rank of an element from given input.
- a. Merge sort algorithm
 - b. **Selection problem** **page 34**
 - c. Brute force technique
 - d. Plane Sweep algorithm
188. How many steps are involved to design the dynamic programming strategy?
- a. 2
 - b. 3
 - c. 1
 - d. **4** **page 92**
189. In Bucket sort, if there are duplicates then each bin can be replaced by a a. Stack

b. Linked list page 69

c. Hash table

d. Heap

190. In merge sort algorithm, we split the array _____ to find index q.

a. from start

b. midway page 28

c. from end

d. both from start or end

191. Find the maximum value of the items which can carry using knapsack Knapsack weight capacity = 50.

Item Weight Value

11070,22020,33080,470 200

a. 280

b. 100

c. 90

200

192. In 2-d maxima problem a point p is said to be dominated by point q if

_____.

a. $p.x \leq q.x$

b. $p.x \leq q.x$ and $p.y \leq q.y$ page 17

c. $p.y \leq q.y$

d. $p.x \geq q.x$ and $p.y \geq q.y$

193. Sorting can be in _____.

- a. Increasing order only
- b. Decreasing order only
- c. Both increasing and decreasing order
- d. Random order

194. Recurrence can be described in terms of _____.

- a. Array
- b. Linear
- c. Tree page 31
- d. Graph

195. The brute-force algorithm for 2D-Maxima runs in order $O(__)$ time.

- a. n
- b. $n(\log n)$
- c. $n*n$ page 18
- d. n^3

196. In plane sweep approach of solving geometric problems, a _____ is swept across the plane.

- a. Line page 18
- b. Plane
- c. Cube
- d. Box

197. Which of the following is calculated with Big Omega notation?

- a. Medium bounds
- b. Upper bounds

c. Lower bounds **Page - 25**

d. Both upper and lower bounds

198. _____ is always based on divide and conquer strategy.

a. Bubble sort

b. Selection sort

c. Pigeon sort

d. Quick sort **page 46**

199. If a matrix has three rows and two columns, then dimensions of matrix will be:

a. 3x2

2x3

b. 3x3

c. 2x2

200. Asymptotic notations are used to describe _____ of an algorithm.

a. Length

b. running time **google**

c. size

d. compile time

201. Catalan numbers are related the number of different _____ on 'n' nodes.

a. Arrays

b. linked lists

c. binary trees **page 85**

d. functions

202. Applying the sieve technique to selection problem, _____ element is picked from array.

- a. Output
- b. Total
- c. Input

d. Pivot **page 35**

203. Dynamic Programming approach is usually useful in solving _____ problems.

- a. Normal

b. Optimization **google**

- c. Array
- d. Loop

204. In recursive formulation of knapsack Problem: $V[0, j] = \underline{\hspace{2cm}}$ for $j \geq 0$

- a. -1

b. 0 **page 93**

- c. 1
- d. 2

205. _____ is a linear time sorting algorithm.

- a. Merge sort

b. Radix sort **page 71**

- c. Quick sort

d. Bubble sort

206. Quick sort is one of the _____ sorting algorithm.

a. Fastest page 19

b. Slowest

c. Major

d. Average

207. The time assumed for each basic operation to execute on RAM model of computation is _____.

a. Infinite

b. Continuous

c. Constant page 10

d. Variable

208. In Sieve Technique, we know the item of interest.
- a. True
 - b. False page 34
209. While analyzing algorithms, _____ and _____ usually considered difficult to calculate.
- a. Finite, Infinite
 - b. Floor, ceiling google
 - c. Row, Column
 - d. Graph, Tree
210. While analysis of the brute-force maxima algorithm, an array sorted in the reverse order is the type of _____ case input.
- a. Best
 - b. Worst page 14
 - c. Somewhat bad
 - d. Average
211. _____ is not useful measure of central tendency of given input set especially when the distribution of values is highly skewed.
- a. Mean
 - b. Mode
 - c. Average
 - d. Median page 34

212. In asymptotical analysis of $n(n-3)$ and $4n*n$, as n becomes large, the dominant (fastest growing) term is some constant times _____.
a. $n+1$
b. $n-1$
c. n
d. $n*n$ page 23
213. In addition to passing in the array itself to Merge Sort algorithm, we will pass in other arguments which are indices.
a. Two P-38
b. Three
c. Four
d. Five

214. In 2d-maximal problem, a point is said to be if it is not dominated by any other point in that space.

a. Member

b. Minimal

c. Maximal

P-11

d. Joint

215. Counting sort assumes that the numbers to be sorted are in the range _____.

a. K to n where n is large

b. 1 to k where k is small

(P-57)

c. K to n where k is small

d. k to n where n is small

216. Insertion sort is an efficient algorithm for sorting a _____ number of elements

a. Large

P-39

b. Small

c. Extra large

d. Medium

217. If the indices passed to merge sort algorithm are _____ then this means that there is only one element to sort.

a. Small page 28

b. Large

c. Equal

d. Not Equal

218. In Knapsack Problem, each item must be entirely accepted or rejected, is called _____ problem.

a. Fractional

b. 0-1 P-92

c. Linear

d. Optimal

219. If the time complexity of an algorithm is $O(n)$. then it is called _____ time complexity.

a. **Linear** **Wikipedia**

b. Constant

c. Average

d. Exponential

220. In the case of _____ analysis does not depend upon on the distribution of input.

a. Merge sort

b. Quick sort **P-50**

c. Insertion sort

d. Heap sort

221. We can use the _____ Property to devise a recursive formulation of the edit distance problem.

a. Small substructure

b. Algorithmic

c. Real

d. **Optimal substructure** **page 78**

222. The following sequence is called _____

1,2,3,5,8,13,21,34,55,.....

a. Optimize sequence

b. **Fibonacci sequence** **page 73**

c. Optimal sequence

d. Overlapping sequence

223. Which one sorting algorithm is best suited to sort an array of 2 million elements?

a. Bubble sort

b. Insert sort

c. Merge sort

d. Quick sort

e. **Ridx Sort** **page 71**

224. We can improve the performance of quick sort if we could be able to __,_____.

a. Skip input elements somehow

b. **Select two or more pivots** **page 34**

c. Skip any sub-array completely

d. Eliminate recursive calls

225. The problem with the brute-force algorithm is that it uses _____ in pruning out de

a. Worst-case time

b. **No intelligence** **page 18**

- c. Outside looping
- d. Artificial intelligence

226. In chain matrix multiplication, the order of the matrices _____.

a. Can be changed

b. Can not be changed page 85

- c. is equal
- d. is reverse

227. In quick sort algorithm, we choose pivot_____.

a. Always the smallest element

b. Greater than 5

c. Randomly page 35

d. Less than 5

228. In Heap Sort algorithm. Heapify procedure is _____ in nature.

a. Recursive

b. Non-Recursive page 43

- c. Fast
- d. Slow

229. When matrix A of 5×3 is multiplied with matrix B of 3×4 then the number of multiplications required will be _____.

- a. 15
- b. 12
- c. 36
- d.

60

230. An algorithm is said to be correct if for every _____ instance, it halts with the correct _____.

a. Input, Output page 13

b. Design, Analysis

c. Value, Key

d. Key, Analysis

231. In chain matrix multiplication, table is filled _____ to find the multiplication of matrix.

a. row wise

b. column wise

c. diagonally

d. bottom-to-up page 86

232. If we have an equation $8n^2 + 7f \cdot n + 5f + 6$ then is large, _____ term will be muchxxxxxxxthe n term and will dominate the running time.

a. $f \cdot g(n)$

b. $g(n) * 2$

c. $n * 2$ page 23

d. $f(n)$

233. For quick sort algorithm. Partitioning takes theta _____.

a. (n)

b. $\log(n)$

c. $n \log(n)$

$n^2 \log(n)$

234. In Heap Sort algorithm, the maximum levels an element can move upward is

a. **Theta ($\log n$)** page 43

b. Big-oh ($\log n$)

c. Omega ($\log n$)

d. $O(1)$ i.e. Constant time

235. _____ programming is essentially recursion without repetition.

a. Fast

b. **Dynamic** page 75

c. Array

d. $n (\log n)$

236. There are no hard formal rules to the syntax of the _____ code.
- a. Basic
 - b. Programming
 - c. Pseudo
 - d. Assembly
237. In Heap Sort algorithm, to remove the maximum element every time.
- a. We call Build-Heap procedure
 - b. Heap Sort algorithm terminates without result
 - c. We call heapify procedure
 - d. Nothing happens
238. Which process is used for avoiding unnecessary repetitions and looking them up again if we need them later.
- a. Greedy Approach
 - b. Memoization page 74
 - c. Divide and conquer
 - d. Recursion
239. The worst-case running time of Quick sort is _____ in order to sort an array of n element.
- a. $O(n \log n)$ page 49
 - b. $O(n)$
 - c. $O(n^2)$
 - d. $O(\log n)$
240. Boolean operation is a _____ operation on an idealized RAM model of computation.
- a. Starting
 - b. Basic page 10

c. Advance

d. Normal

241. In chain matrix multiplication, if there are n items, there are _____ ways in which outer most pair of parentheses can placed.

a. n^2

b. $2n$

c. $n+1$

d. $n-1$ page 85

242. The number of nodes in a complete binary tree of height h is: a. $2 * (h+1) - 1$

b. $2^{(h+1)} - 1$ page 40

c. $2 * (h+1)$

d. $((h+1)^2) - 1$

Cs502 Solved MCQS for Final Term

Including Current Quizzes

July 2013

Question

Bellman Ford Algorithm applies relaxations to every _____

Vertex of ...

Edge of Graph and repeats this $V-1$ times (p159)

Question

Floyd-Warshall Algorithm is _____

Based on dynamic programming (p161)

Question

Running time of Floyd-Warshall Algorithm is _____

Θn^2

$$\Theta n^4$$

$$\Theta n^2 \quad (\text{p164})$$

$$3^n$$

Question

Space used by Floyd-Warshall Algorithm is _____

$$\Theta n^2 \quad (\text{p164})$$

$$\Theta n^4$$

$$\Theta n^3$$

$$3^n$$

Question

Huffman encoding Algorithm is a _____

Greedy Algorithm (p99)

Dynamic and Greedy

Question

Graphs cannot be traversed by brute-force technique.

True (p117) False

Question

Brute-force technique is always optimal for Graph traversal.

True **False** (p117)

Question

For graph traversal, Breadth-first search strategy _____

is always recursive

Cannot be recursive

Cannot be non-recursive

Can be both recursive and non-recursive

Question #

Greedy algorithm can do very poorly for some problems.

True (p97)

False

Question #

A greedy algorithm sometimes works well for optimization problems.

True (p97)

False

Question #

A greedy algorithm does not work in phases.

True

False (p97)

Question #

Greedy algorithm approach solves both 0/1 Knapsack and Fractional Knapsack problems.

True

False

Question #

Dynamic Programming approach solves both 0/1 Knapsack and Fractional Knapsack problems.

True

False

Question #

In _____ based solution of knapsack problem, we consider 2 cases. Leave object Or Take object.

Brute force

Dynamic programming (p93)

Question #

In brute force based solution of knapsack problem, we consider 2 cases. Leave object Or Take object.

True

False (p93)

Question #

In DP based solution of knapsack problem, $V[0, j] = \underline{\hspace{2cm}}$ if $j \geq 0$

-1

0 (p93)

1

2

Question

In DP based solution of knapsack problem, $V[0, j] = 1$ if $j \geq 0$ True

False (p93)

Question

In DP based solution of knapsack problem, to compute entries of V we will imply a/an _____ approach.

Subjective

Inductive (p93)

Brute force

Combination

Question

In DP based solution of knapsack problem, if we decide to take an object i, then we gain a value of _____.

w_{i-1}

v_{i-1}

w_i

vi (p93)

Question #

In Fractional Knapsack problem, one is allowed to take fraction of an item for _____

for fraction of the weight

for fraction of the value

Both, fraction of the weight and value (p109)

None

Question #

_____ approach is optimal for the fractional knapsack problem.

Divide and Conquer

Dynamic Programming

Greedy algorithm (p110)

Brute force

Question #

In ____ Knapsack Problem, limitation is that an item can either be put in the bag or not-fractional items are not allowed.

0 1 **0/1** Fractional

Question #

In Fractional Knapsack problem, the goal is to _____

minimize the value of items without exceeding the total weight limit of W

maximize the value of items without exceeding the total weight limit of W (p109)

minimize the value of items even if exceeding the total weight limit of W

maximize the value of items even if exceeding the total weight limit of W

Question #

In solving Fractional Knapsack problem, we do not need sorting.

True

False (p109)

Question

Suppose that a graph $G = (V, E)$ is implemented using adjacency lists. What is the complexity of a breadth-first traversal of G ? Select correct option:

$O(|V|^2)$

$O(|V| + |E|)$

$O(|V|^2|E|)$

$O(|V| + |E|)$ (p116)

Question

For traversing graphs, Breadth-first search can be visualized as a wave front propagating inwards towards root (or source) node.

True

False (p117)

Question

Breadth-first search is **not** a traversal strategy for graphs.

True

False (p119)

Question

Breadth-first search is **not** a popular algorithm technique used for traversing graphs.

True

False (p119)

Question

Breadth-first search begins at a root node and inspects all the nodes **except** neighboring ones.

True

False (117)

Question

Each time we traverse graph by Breath-first search algorithm, we count the distance from _____

Starting node (p117)

neighbors of the starting node

right most node

left most node

Question

Suppose we are given a set $S = \{a_1, a_2, \dots, a_n\}$ of n activities that are to be scheduled to use some resource. Each activity a_i must be started at a given start time s_i and ends at a given finish time f_i . We want to schedule the largest number of activities on the resource. It is an example of _____. Searching

Dynamic Programming

Activity selection (p105)

Divide and Conquer

Question

In general, the Activity selection problem is to select a _____

minimum-size set of interfering activities

maximum-size set of mutually non-interfering activities (p105)

maximum-size set of interfering activities

minimum-size set of mutually non-interfering activities

Question #

In Activity selection (using Greedy approach), intuitively _____

Short activities are not preferable

There are always short activities as input

We do not like long activities (p105)

It does not matter about the length of activities

Question #

The activity schedule is a simple scheduling problem for which the ____ approach provides an optimal solution.

Dynamic Programming

Graph

Divide and Conquer

Greedy Algorithm (p105)

Question #

The activity scheduling is a simple scheduling problem for which the greedy algorithm approach provides a/an _____ solution.

Simple

Sub optimal

Optimal (p105)

Non optimal

Question #

A number of lectures are to be given in a single lecture hall. Optimum scheduling for this is an example of Activity selection.

True (p105)

False

Question #

In Activity scheduling algorithm, each activity is represented by a _____

Circle

Square

Triangle

Rectangle (p106)

Question #

In Activity scheduling algorithm, the width of a rectangle _____

Is always ignored

Directs towards recursion

Should be maximized

Indicates the duration of an activity (p106)

Question #

In Activity scheduling algorithm, the time is dominated by sorting of the activities by _____

start times

finish times (p106)

Question #

The complexity of Activity scheduling algorithm is _____

$O(n)$

$O(\log n)$

$O(n \log n)$ (p106)

$O(n^2)$

Question #

In Activity scheduling algorithm, as base case if there are no activities then Greedy algorithm _____

cannot be optimized

is solved using Recursion

is transformed into Dynamic Programming

is trivially optimal (p109)

Question #

Bag is a _____

type of algorithm

data structure (p119)

program

compiler

Question

For _____ problems, greedy approach always gets optimum.

Select correct option:

Two

Some (p97)

All

None of the

Question

Fixed-length codes may not be efficient from the perspective of _____ the total quantity of data.

Minimizing (p99)

Averaging

Maximizing

Summing

Question

There are _____ nested loops in DP based algorithm for computing the minimum cost of chain matrix multiplication.

2

3 (p90)

4

5

Question

Time complexity of DP based algorithm for computing the minimum cost of chain matrix Multiplication is ----.

$\log n$ n n^2 **n^3 (p90)**

Question

In greedy algorithm, at each phase, you take the _____ you can get right now, without regard for future consequences.

Worst Minimum Good **Best (p97)**

Question #

An optimization problem is one in which you want to find, not just a solution, but the _____ solution.

Simple Good **Best (p97)** Worst

Question #

In _____ algorithm, you hope that by choosing a local optimum at each step, you will end up at a global optimum.

Simple Non-Greedy. **Greedy (p97)** Brute force

Question #

The string |xyz|, if coded with ASCII code, the message length would be 24 bits.

True False

Question #

If the string |lmncde| is coded with ASCII code, the message length would be --- bits.

24 36 **48** 60

Question #

If there are n items, there are _____ possible combinations of the items.

2 n 2^n (p92) 3^n

Question #

In Huffman encoding, for a given message string, the frequency of occurrence (relative probability) of each character in the message is determined last.

True **False** (p100)

Question #

In DP based solution of knapsack problem, to compute entries of V we will imply a/an _____ approach.

Subjective **Inductive** (p93) Brute force Combination

Question #

Inductive approach to compute entries of V is implied in _____ based solution of knapsack problem.

Brute force **Dynamic programming** (p93)

Question #

In brute force based solution of knapsack problem, we consider 2 cases, Leave object Or Take object.

True (p93) False

Question #

There are _____ nested loops in DP based algorithm for computing the minimum cost of chain matrix multiplication.

2

3

4

5 (not confirm page 90)

Question #

Those problems in which Greedy finds good, but not always best is called a greedy_____.

Algorithm

Solution

Heuristic (p97)

Result

Question #

The Huffman codes provide a method of _____ data efficiently.

Reading

Encoding (p99)

Decoding

Printing

Question #

Fixed-length codes are known for easy break up of a string into its individual characters.

True (p99) False

Question #

The Huffman codes provide a method of _____ data efficiently.

Reading **Encoding (p99)** Decoding Printing

Question #

In Huffman encoding, the _____ is the number of occurrence of a character divided by the total characters in the message.

Counting

Parsing

Probability (p100)

Weight

Question #

In Huffman encoding, for a given message string, the frequency of occurrence (relative probability) of each _____ in the message is done first.

Word

Space

Digit

Character (p100)

Question #

The Huffman algorithm finds a (n) _____ solution.

Optimal (Reference)

Non-optimal

Exponential

Polynomial

Question #

The Huffman algorithm finds an **exponential** solution True

False (Reference)

Question #

The Huffman algorithm finds a **polynomial** solution True

False (Reference)

Prefix Property:

The codewords assigned to characters by the Huffman algorithm have the property that no codeword is a prefix of any other:

Question #

Non-optimal or greedy algorithm for money change takes _____

O(k) (p99)

O(kN)

O(2k)

O(N)

Question #

The Huffman codes provide a method of encoding data **inefficiently** when coded using ASCII standard.

True **False (p99)**

- The Huffman codes provide a method of encoding data **efficiently**.

Question

The codeword assigned to characters by the Huffman algorithm have the property that no codeword is the **postfix** of any other. True **False**

Question

The codeword assigned to characters by the Huffman algorithm have the property that no codeword is the **prefix** of any other. **True (p101)** False

Question

Using ASCII standard the string “abacdaacacwe” will be encoded with _____ bits

Select correct option:

64

128

96 (p101 12x8=96)

120

Question

Using ASCII standard the string abacdaacac will be encoded with _____ bits.

80 (p99) 160 320 100

- Consider the string “ abacdaacac”. if the string is coded with ASCII codes, the message length would be
- **$10 \times 8 = 80$ bits.**

Question

Using ASCII standard the string abacdaacac will be encoded with 160 bits. True **False (p99)**

Question

Using ASCII standard the string abacdaacac will be encoded with 320 bits. True **False (p99)**

Question

Using ASCII standard the string abacdaacac will be encoded with 100 bits. True **False (p99)**

Question

Using ASCII standard the string abacdaacac will be encoded with 32 bytes. True **False (p99)**

Question

The greedy part of the Huffman encoding algorithm is to first find two nodes with **smallest** frequency.

True (p100) False

Question

The greedy part of the Huffman encoding algorithm is to first find two nodes with **character** frequency.

True **False (p100)**

Question

The greedy part of the Huffman encoding algorithm is to first find two nodes with **larger** frequency.

True **False**

Question

Huffman algorithm uses a greedy approach to generate an **antefix code** T that minimizes the expected length $B(T)$ of the encoded string. True **False (p102)**

Question

Huffman algorithm uses a greedy approach to generate a postfix code T that minimizes the expected length $B(T)$ of the encoded string. True **False (p102)**

Question

Huffman algorithm uses a greedy approach to generate a prefix code T that minimizes the expected length $B(T)$ of the encoded string. **True (p102)** False

Question

Huffman algorithm generates an optimum prefix code.

True (p102) False

Question

A graph is said to acyclic if it contains _____

at least one cycle

exactly one cycle

always more than one cycle

no cycle (p116)

Question

Graph is **not** a good way to model some sort of "connection" or "relationship" or "interaction" between pairs of objects taken from a set of objects.

True **False** (p113)

Question

A graph is not connected if every vertex can reach every other vertex.

True **False** (p116)

Question

Traversing a graph means visiting _____ in the graph.

no node

at least one

more than one node

every node (p119)

Question

Graphs are important _____ model for many application problems.

Unsystematic

Mathematical (p113)

Haphazard

Unpredictable

Question

In a digraph, the number of edges coming in of a vertex is **not** called the in-degree of that vertex.

True **False** (p114)

Question

In a digraph, the number of edges coming out of a vertex is **not** called the out-degree of that vertex.

True **False (p114)**

Question

For a digraph $G = (V, E)$, Sum of in-degree(v) _____

Not equal to Sum of out-degree(v)

= Sum of out-degree(v) (p115)

< Sum of out-degree(v)

> Sum of out-degree(v)

Question

Graphs **cannot be** represented by an adjacency matrix.

True **False (p116)**

Question

A vertex **a** is **not** adjacent to vertex **b** if there is an edge from a to b.

True **False (p113)**

Question

A **dense undirected** graph is:

Select correct option:

A graph in which $E = O(V^2)$ (Reference)

A graph in which $E = O(V)$

A graph in which $E = O(\log V)$

All items above may be used to characterize a dense undirected graph

Question

What algorithm technique is used in the implementation of **Kruskal solution** for the MST?

Greedy Technique (p142)

Divide-and-Conquer Technique

Dynamic Programming Technique

The algorithm combines more than one of the above techniques

Question

A digraph is strongly connected under what condition?

A digraph is strongly connected if for every pair of vertices $u, v \in V$, u can reach v .

A digraph is strongly connected if for every pair of vertices $u, v \in V$, u can reach v and vice versa.

(p135) A digraph is strongly connected if for at least one pair of vertex $u, v \in V$, u can reach v and vice versa.

A digraph is strongly connected if at least one third pair of vertices $u, v \in V$, u can reach v and vice versa.

Question

The relationship between number of back edges and number of cycles in DFS is,

Both are equal

Back edges are half of cycles

Back edges are one quarter of cycles

There is no relationship between no. of edges and cycles (p131)

Question

Back edge is?

(u, v) where v is an ancestor of u in the tree. (p128)

(u,v) where u is an ancestor of v in the tree.

(u, v) where v is an predecessor of u in the tree.

None of above

Question

Forward edge is?

(u, v) where u is a proper descendent of v in the tree.

(u, v) where v is a proper descendent of u in the tree. (p129)

(u, v) where v is a proper ancestor of u in the tree.

(u, v) where u is a proper ancestor of v in the tree.

Question

Edge (u, v) is a forward edge if

u is a proper descendant of v in the tree

v is a proper descendant of u in the tree

None of these

Question

Cross edge is?

(u, v) where u and v are not ancestor of one another

(u, v) where u is ancestor of v and v is not descendent of u.

(u, v) where u and v are not ancestor or descendent of one another (p129)

(u, v) where u and v are either ancestor or descendent of one another.

Question

If you find yourself in maze the better traversal approach will be?

Select correct option:

BFS

BFS and DFS both are valid (p119)

Level order

DFS

Question

In digraph $G=(V,E)$; G has cycle if and only if

Select correct option:

The DFS forest has forward edge.

The DFS forest has back edge (p131)

The DFS forest has both back and forward edge

BFS forest has forward edge

Question

What is generally true of Adjacency List and Adjacency Matrix representations of graphs?

Select correct option:

Lists require less space than matrices but take longer to find the weight of an edge (v1,v2)

Lists require less space than matrices and they are faster to find the weight of an edge (v1, v2)

Lists require more space than matrices and they take longer to find the weight of an edge (v1, v2)

Lists require more space than matrices but are faster to find the weight of an edge (v1, v2)

Question

There is relationship between number of **back edges** and **number of cycles** in DFS

Select correct option:

Both are equal.

Cycles are half of back edges.

Cycles are one fourth of back edges.

There is no relationship between back edges and number of cycle (p131)

Question #

You have an adjacency list for G, what is the time complexity to compute Graph transpose G^T ?

Select correct option:

(V+E) (Reference)

V.E

V

E

Question #

Although it requires more complicated data structures, **Prim's algorithm** for a minimum spanning tree is **better** than **Kruskal's** when the graph has a **large number** of vertices.

True False

Question #

Maximum number of vertices in a Directed Graph may be $|V|^2$

True False

Question

Dijkstra's algorithm:

Has greedy approach to find all shortest paths

Has both greedy and dynamic approach to find all shortest paths

Has greedy approach to compute single source shortest paths to all other vertices (p154) Has both greedy and dynamic approach to compute single source shortest paths to all other vertices.

Question

Dijkstra's single source shortest path algorithm works if all edges weights are **non-negative** and **there are no negative** cost cycles. True **False**

Question

Dijkstra's single source shortest path algorithm works if all edges weights are **non-negative** and **there are no negative** cost cycles. **True (p159)** False

Question

Dijkstra's single source shortest path algorithm works if all edges weights are **negative** and **there are no negative** cost cycles.

True **False**

Question

Dijkstra's algorithm is operates by maintaining a subset of vertices.

True False

Question

Floyd-Warshall algorithm is a dynamic programming algorithm; the genius of the algorithm is in the clever recursive formulation of the shortest path problem. **True (p162)** False

Question

Floyd-Warshall algorithm, as in the case with DP algorithms, we avoid recursive evaluation by generating a table for $d_{ij}^{(k)}$

True (p164)

False

Question #

The term coloring came from the original application which was in **map drawing**.

True (p173)

False

Question #

The term “coloring” came from the original application which was in **architectural design**. True

False

Question #

In the **clique cover** problem, for two vertices to be in the same group, they must be _____ each other.

Apart from Far from Near to **Adjacent to (p176)**

Question #

In the clique cover problem, for two vertices to be in the same group, they **must be adjacent** to each other.

True (p176)

False

Question #

In the **clique cover** problem, for two vertices to be in the same group, they **must be apart** from each other.

True **False (p176)**

Question

The difference between Prim's algorithm and Dijkstra's algorithm is that Dijkstra's algorithm uses a different key.

True (p156)

False

Question

The difference between Prim's algorithm and Dijkstra's algorithm is that Dijkstra's algorithm uses a same key.

True **False (p156)**

Question

You have an adjacency list for G , what is the time complexity to compute graph transpose G^T ?

$(V+E)$ (p138)

Question

Given an adjacency list for G , it is possible to compute G^T in $\Theta(V + E)$ time.

It takes $O(\log V)$ to extract a vertex from the priority queue.

Question

Overall time for Kruskal is $\Theta(E \log E) = \Theta(E \log V)$ if the graph is sparse. **True (p149)**

Question

An optimization problem is one in which you want to find,

Not a solution

An algorithm

Good solution

The best solution (p97)

Question

Although it requires more complicated data structures, Prim's algorithm for a minimum spanning tree is better than Kruskal's when the graph has a large number of vertices.

True

False

Question

If a graph has v vertices and e edges then to obtain a spanning tree we have to delete v edges.

$v - e + 5$ edges

$v + e$ edges.

None of these

Question

Bellman-Ford allows negative weights edges and negative cost cycles.

True

False (p159)

- Bellman-Ford allows negative weights edges and no negative cost cycles.

Question

We do sorting to,

keep elements in random positions

keep the algorithm run in linear order

keep the algorithm run in $(\log n)$ order

keep elements in increasing or decreasing order (p39)

Question

After partitioning array in Quick sort, pivot is placed in a position such that

Values smaller than pivot are on left and larger than pivot are on right (Ref)

Values larger than pivot are on left and smaller than pivot are on right

Pivot is the first element of array

Pivot is the last element of array

Question

Dynamic programming algorithms need to store the results of intermediate sub-problems.

True False

Question

Which statement is true?

If a dynamic-programming problem satisfies the optimal-substructure property, then a locally optimal solution is globally optimal.

If a greedy choice property satisfies the optimal-substructure property, then a locally optimal solution is globally optimal.

Both of above None of above

Question

What **general property** of the list indicates that the graph has an **isolated vertex**?

There is Null pointer at the end of list.

The Isolated vertex is not handled in list.

Only one value is entered in the list.

There is at least one null list.

Question

Depth first search is **shortest path** algorithm that works on **un-weighted** graphs.

True **False (p153)**

- The **breadth-first-search** algorithm is a shortest-path algorithm that works on **un-weighted graphs**.

Question

Which is true statement?

Select correct option:

Breadth first search is shortest path algorithm that works on un-weighted graphs (p153)

Depth first search is shortest path algorithm that works on un-weighted graphs.

Both of above are true.

None of above are true.

Question

Shortest path problems can be solved efficiently by modeling the road map as a graph.

True False

Question

Which is true statement in the following?

Kruskal algorithm is multiple source technique for finding MST.

Kruskal's algorithm is used to find minimum spanning tree of a graph, time complexity of this algorithm is $O(EV)$ Both of above

Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best tree edge) when the graph has relatively few edges.

Question

Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best tree edge) when the graph has relatively few edges.

True False

Question

What is the time complexity to **extract a vertex** from the **priority queue** in **Prim's algorithm**?

Select correct option:

$O(\log E)$

$O(V)$

$O(V+E)$

$O(\log V)$ (p152)

Question

In Prim's algorithm, the additional information maintained by the algorithm is the **length of the shortest edge** from vertex v to points already in the tree. 

a) True **b) False** c) unknown

Question

Although it requires more complicated data structures, Prim's algorithm for a minimum spanning tree is better than Kruskal's when the graph has a **large number of vertices**.

a) True b) False c) unknown

Question

Which statement is true?

- ▶ The running time of Bellman-Ford algorithm is $T(VE)$
- ▶ Both Dijkstra's algorithm and Bellman-Ford are based on performing **repeated relaxations**
- ▶ The 0-1 knapsack problem is hard to solve

Only i Only iii Both i and iii **All of these**

Question

Which of the following arrays represent **descending (max) heaps**?

i. [10,7,7,2,4,6] ii. [10,7,6,2,4,7]

iii. [10,6,7,2,4,6] iv. [6,6,7,2,4,10]

Only ii

Only iv

Both ii and iv

Both i and iii

Question

Which of the following statement(s) is/are correct?

(a) $O(n \log n + n^2) = O(n^2)$.

(b) $O(n \log n + n^2) = O(n^2 \log 2n)$

(c) $O(c n^2) = O(n^2)$ where c is a constant.

(d) $O(c n^2) = O(c)$ where c is a constant.

(e) $O(c) = O(1)$ where c is a constant.

Only (a) & (e)

Both (c) and (e)

Question

Which of the shortest path algorithms would be most appropriate for finding paths in the graph with **negative edge weights** and **cycles**?

► Dijkstra's Algorithm

► Bellman-Ford Algorithm.

► Floyd Warshall Algorithm

Only ii

Only iii

Both ii & iii

Question

If, in a DFS forest of digraph $G = (V, E)$, $f[u] = f[v]$ for an edge $(u, v) \in E$ then the edge is called

Back edge

Forward edge

Cross Edge

Tree Edge

None of these

Question

Best and worst case times of an algorithm may be same. **True** False

Question

Can an adjacency matrix for a directed graph ever not be square in shape? **Yes** No

Question

Consider the following Huffman Tree. The binary code for the string **TEA** is:

10 00 010

011 00 010

10 00 110

11 10 110

String Encoding	
TEA	10 00 010
SEA	011 00 010
TEN	10 00 110

Question

Can an adjacency matrix for a directed graph ever not be square in shape? Yes **No**

- No. since we want to describe the relationship between each node and each other node, we need precisely n^2 matrix entries.

Question

Suppose we have two problems A and B .Problem A is polynomial-time reducible and problem B is NP-complete. If we reduce problem A into B then problem A becomes NP-complete **Yes** No

Question

If a problem is NP-complete, it must also be in NP.

a) True b) False c) unknown

Question

If a problem is in NP, it must also be in **P**.

True **False (p178)** unknown

Question

If a problem is in NP-complete, it must also be in **NP**.

True False

Cs502 Solved MCQS for MidTerm

Including Current Quizes

26 May 2013

Question # 1

Word Algorithm comes from the name of the Muslim author:

Abu Ja'far Mohammad ibn Musa al-Khowarizmi. (p7)

Question # 2

Al-Khwarizmi's work was written in a book titled

al Kitab al-mukhtasar fi hisab al-jabr wa'l-muqabalah (p7)

Question # 3

Analysis of Selection algorithm ends up with,

Select correct option:

T(n) (p37) $T(1 / 1 + n)$ $T(n / 2)$ $T((n / 2) + n)$

Question # 4

The function $f(n) = n(\log n + 1)/2$ is asymptotically equivalent to $n \log n$. Here Upper Bound means the function $f(n)$ grows asymptotically _____ faster than $n \log n$.

Select correct option:

More Quiet **Not (p24)** At least

Question # 5

The running time of an algorithm would not depend upon the optimization by the compiler but that of an implementation of the algorithm would depend on it. Select correct option:

True **False (p13)**

Question # 6

After sorting in merge sort algorithm, merging process is invoked.

Select correct option:

True (p27)

False

Question # 7

Asymptotic growth rate of the function is taken over _____ case running time.

Select correct option:

Best

Average

Worst (p14)

Normal

Question # 8

In analysis of $f(n)=n(n/5)+n-10 \log n$, $f(n)$ is asymptotically equivalent to _____.

n

$2n$

$n+1$

n^2 (p23)

Question # 9

$f(n)$ and $g(n)$ are asymptotically equivalent. This means that they have essentially the same _____ for large n .

Results

Variables

Size

Growth rates (p23)

Question # 10

Algorithm is concerned with _____ issues.

Macro

Micro 1

Both Macro & Micro (p8)

Normal

Question # 11

We cannot make any significant improvement in the running time which is better than that of brute-force algorithm.

True

False (p18, p92)

Question # 12

In addition to passing in the array itself to Merge Sort algorithm, we will pass in _____ other arguments which are indices.

Two (p28)

Three

Four

Five

Question # 13

For the worst-case running time analysis, the nested loop structure containing one “for” and one “while” loop, might be expressed as a pair of _____ nested summations.

1

2 (p14)

3

4

Question # 14

The O-notation is used to state only the asymptotic _____ bounds.

Two

Lower

Upper (p25)

Both lower & upper

Question # 15

Which of the following is calculated with **big O notation**?

Lower bounds

Upper bounds (p25)

Both upper and lower bound

Medium bounds

Question # 16

The Ω -notation allows us to state only the asymptotic _____ bounds.

Lower (p25) Upper

Both upper and lower bound Medium bounds

Question # 17

The time assumed for each basic operation to execute on RAM model of computation is _____

Infinite Continuous **Constant** (p10) Variable

Question # 18A

If the indices passed to merge sort algorithm are not equal, the algorithm may return immediately.

True **False** (p28)

Question #18B

If the indices passed to merge sort algorithm are _____, then this means that there is only one element to sort.

Small Large **Equal** (p28) Not Equal

Question # 19

Brute-force algorithm uses no intelligence in pruning out decisions.

True (p18) False

Question # 20

In analysis, the **Lower Bound** means the function grows asymptotically **at least** as fast as its largest term.

True (p24) False

Question # 21

In analysis, the **Upper Bound** means the function grows asymptotically **no faster** than its largest term.

True (p24) False

Question # 22

For small values of n, any algorithm is fast enough. Running time does become an issue when n gets large.

True (p14) Fast

Question # 23

The array to be sorted is not passed as argument to the merge sort algorithm.

True **False (p28)**

Question # 24

In simple brute-force algorithm, we give no thought to efficiency.

True (p11) False

Question # 25

The ancient Roman politicians understood an important principle of good algorithm design that is plan-sweep algorithm.

True **False (p27) [Divide and Conquer]**

Question # 26

In 2d-space a point is said to be _____ if it is not dominated by any other point in that space.

Select correct option:

Member Minimal **Maximal** **(p11)** Joint

Question # 27

An algorithm is a mathematical entity that is dependent on a specific programming language.

Select correct option:

True **False** **(p7)**

Question # 28

Fibonacci sequence was posted by _____

Leonardo Pisano **(p73)**

Question # 29

$8n^2 + 2n - 3$ will eventually exceed $c_2(n)$ no matter how large we make c_2 .

True **(p25)** False

Question # 30

If we associate (x, y) integers pair to cars where x is the speed of the car and y is the negation of the price. High y value for a car means a _____ car.

Fast Slow Expensive **Cheap** **(p11)**

Question # 31

The running time of quick sort depends heavily on the selection of:

No of inputs Arrangement of elements in array

Size of elements **Pivot element** (p49)

Question # 32

Which sorting algorithm is faster?

$O(n^2)$ $O(n \log n)$ **$O(n+k)$** (p58) $O(n^3)$

Question # 33

Q knapsack problem is called a “0-1” problem, because

Each item must be entirely accepted or rejected (p92)

Question # 34

Which statement is true?

Select correct option:

If a **dynamic-programming problem** satisfies the optimal-substructure property, then a **locally** optimal solution is globally optimal.

If a **greedy choice property** satisfies the optimal-substructure property, then a locally optimal solution is globally optimal.

Both of above (p77, p98) None of above

Question # 35

Merge sort is stable sort, but not an in-place algorithm

True (p54) False

Question # 36

In counting sort, once we know the ranks, we simply _____ numbers to their final positions in an output array.

Delete **copy (p57)** Mark arrange

Question # 37

Dynamic programming algorithms need to store the results of intermediate sub-problems. **True (p75)** False

Question # 38

A $p \times q$ matrix A can be multiplied with a $q \times r$ matrix B. The result will be a $p \times r$ matrix C. There are $(p \cdot r)$ total entries in C and each takes _____ to compute.

$O(q)$ **$O(1)$ (p84)** $O(n^2)$ $O(n^3)$

Question # 39

Due to left-complete nature of binary tree, heaps can be stored in

Link list Structure **Array (p40)** None of above

Question # 40

Efficient algorithm requires less computational _____

Memory Running Time **Memory and Running Time (p9)** Energy

Question # 41

Merge sort makes two recursive calls. Which statement is true after these recursive calls finish, but before the merge step?

The array elements form a heap

Elements in each half of the array are sorted amongst themselves

Elements in the first half of the array are less than or equal to elements in the second half of the array

None of the above

Question # 42

Who invented Quick sort procedure?

Hoare Sedgewick Mellroy Coreman

Question # 43

Merge sort requires extra array storage. **True (p54)** False

- Mergesort is a stable algorithm but **not** an in-place algorithm. It requires extra array storage.

Question # 44

In which order we can sort?

Increasing order only decreasing order only

Increasing order or decreasing order both at the same time

Question # 45

In the analysis of **Selection algorithm**, we make a number of passes, in fact it could be as many as,

$T(n)$ $T(n/2)$ **$\log n$ (p37)** $n/2 + n/4$

Question # 46

How much time **merge sort** takes for an array of numbers?

$T(n^2)$ $T(n)$ $T(\log n)$ **$T(n \log n)$ (p40)**

Question # 47A

Sieve Technique applies to problems where we are interested in finding a **single** item from a larger set of ____

n items (p34) phases pointers constant

Question # 47B

The sieve technique works in phases as follows. It applies to problems where we are interested in finding

a **single** item from a larger set of n items. **(p34)**

Question # 47C

The sieve technique works where we have to find _____ item(s) from a large input.

Single (p34)

Two

Three

Similar

Question # 48

The sieve technique works in _____ as follows

Phases (p34)

numbers

integers

routines

Question # 49

For the heap sort, access to nodes involves simple _____ operations.

Arithmetic (p41)

binary

algebraic

logarithmic

Question # 50

The analysis of Selection algorithm shows the total running time is indeed _____ in n ,

Arithmetic

geometric

linear (p39)

orthogonal

Question # 51

Slow sorting algorithms run in,

$T(n^2)$ (p39)

$T(n)$

$T(\log n)$

$T(n \log n)$

Question # 52

A heap is a left-complete binary tree that conforms to the

Increasing order only decreasing order only **heap order** (log n) order

Question # 53

For the heap sort we store the tree nodes in

Level-order traversal (p40) in-order traversal pre-order traversal post-order traversal

Question # 54A

In Sieve Technique we do not know which item is of interest **True (p34)** False

Question # 54B

In Sieve Technique, we know the item of interest.

True **False (p34)**

Question # 55

Divide-and-conquer as breaking the problem into a small number of

Pivot Sieve **smaller sub problems (p34)** Selection

Question # 56

For the sieve technique we solve the problem,

Recursively (p34) mathematically precisely accurately

Question # 57

The reason for introducing **Sieve Technique** algorithm is that it illustrates a very important special case of, **Divide-and-conquer (p34)** decrease and conquer greedy nature 2-dimension Maxima

Question # 58

Sorting is one of the few problems where provable _____ bounds exists on how fast we can sort,

Upper **Lower (p39)** Average log n

Question # 59

Sieve Technique can be applied to selection problem? **True (p35)** False

Question # 60

One of the clever aspects of heaps is that they can be stored in arrays without using any_____

Pointers (p40)

constants variables functions

Question # 61

Heaps can be stored in arrays without using any pointers; this is due to the ____ nature of the binary tree,

Left-complete (p40)

right-complete tree nodes tree leaves

Question # 62

How many elements do we eliminate in each time for the Analysis of Selection algorithm?

n / 2 elements (p36)

(n / 2) + n elements n / 4 elements 2 n elements

Question # 63

The sieve technique is a special case, where the number of sub problems is just

5 Many

1 (p34)

few

Question # 64

For the Sieve Technique we take time

$T(nk)$

(p34)

$T(n / 3)$

n^2

$n/3$

Question # 65

The number of nodes in a complete binary tree of height h is

Select correct option:

$2^{(h+1)} - 1$

$2 * (h+1) - 1$

$2 * (h+1)$

$((h+1) ^ 2) - 1$

Question # 66

Random access machine or RAM is a/an _____

Machine build by Al-Khwarizmi

Mechanical machine

Electronics machine

Mathematical model (p10)

Question # 67

Analysis of Selection algorithm ends up with,

$T(n)$

$T(1 / 1 + n)$

$T(n / 2)$

$T((n / 2) + n)$

Question # 68

Continuation sort is suitable to sort the elements in range 1 to k

1. K is Large 2. K is not known 3. K may be small or large

4. K is small (p57)

Question # 69

Counting sort is suitable to sort the elements in range 1 to k:

K is large

K is small (p57)

K may be large or small

None

Question # 70

Continuing sort has time complexity of?

1. $O(n)$ (p58) 2. $O(n+k)$ 3. $O(n \log n)$

Question # 71

Counting sort has time complexity:

- $O(n)$ (p58) $O(n+k)$ $O(k)$ $O(n \log n)$

Question # 72

Memoization is _____:

-To store previous results for further use.

-To avoid unnecessary repetitions by writing down the results of recursive calls and looking them again if needed later. (p74)

-To make the process accurate -None of the above

Question # 73

In Quick sort algorithm, constants hidden in $T(n \lg n)$ are:

- Large Medium Not known **Small (Ref)**

Question # 74

Quick sort is based on divide and conquer paradigm; we divide the problem on base of pivot element and:

There is explicit combine process as well to conquer the solution.

No work is needed to combine the sub-arrays, the array is already sorted

Merging the subarrays None of above.

Question # 75

In RAM model instructions are executed _____

In Random Access Machine, instructions are executed _____

One after another (p10) Parallel Concurrent Random

Question # 76

In the analysis of Selection algorithm, we eliminate a constant fraction of the array with each phase; we get the

convergent _____ series in the analysis,

linear arithmetic **geometric (p37)** exponent

Question # 77

_____ is a graphical representation of an algorithm

Σ notation Θ notation **Flowchart (Ref)** Asymptotic notation

Question # 78

A RAM is an idealized machine with _____ random-access memory.

256MB 512MB **An infinitely large (p10)** 100GB

Question # 79

What type of instructions Random Access Machine (RAM) can execute? Choose best answer

Algebraic and logic Geometric and arithmetic
Arithmetic and logic (p10) Parallel and recursive

Question # 80

What will be the total number of **max comparisons** if we run **brute-force maxima** algorithm with n elements?

n^2 (p14) $2n/n$ n 8n

Question # 81

What is the solution to the recurrence $T(n) = T(n/2) + n$.

$O(\log n)$

$O(n)$

$O(n \log n)$

$O(n^2)$

Question # 82

Consider the following code:

```
For(j=1; j<n; j++)  
    For(k=1; k<15; k++)  
        For(l=5; l<n; l++)  
        {  
            Do_something_constant();  
        }
```

What is the order of execution for this code.

$O(n)$

$O(n^3)$

$O(n^2 \log n)$

$O(n^2)$

Question # 83

Consider the following Algorithm: Fun(n){ if (n=1) return 1 else return (n * Fun(n-1)) } Recurrence for the above algorithm is:

$$nT(n-1)+1$$

$$2T(n-1)+1$$

$$T(n-1)+cn$$

$$T(n-1)+1$$

Question # 84

Consider the following Algorithm:

Factorial (n){

if (n=1)

return 1

else

return (n * Factorial(n-1))

{

Recurrence for the following algorithm is:

$$T(n) = T(n-1) + 1$$

$$T(n) = nT(n-1) + 1$$

$$T(n) = T(n-1) + n$$

$$T(n) = T(n(n-1)) + 1$$

Question # 85

What is the total time to heapify?

$$O(\log n)$$

(p43)

$$O(n \log n)$$

$$O(n^2 \log n)$$

$$O(\log^2 n)$$

Question # 86

When we call heapify then at each level the comparison performed takes time

$$\text{It will take } \Theta(1)$$

(p43)

Time will vary according to the nature of input data

It cannot be predicted

It will take $\Theta(\log n)$

Question # 87

In Quick sort, we don't have the control over the sizes of recursive calls

-True (p49) -False -Less information to decide -Either true or false

Question # 88

Is it possible to sort without making comparisons? **Yes (p57)** No

Question # 89

If there are $\Theta(n^2)$ entries in **edit distance matrix** then the total running time is

$\Theta(1)$ **$\Theta(n^4)$ (p84)** $\Theta(n)$ $\Theta(n \log n)$

Question # 90

For **Chain Matrix Multiplication** we cannot use divide and conquer approach because,

We do not know the optimum k (p86)

We use divide and conquer for sorting only

We can easily perform it in linear time

Size of data is not given

Question # 91

The Knapsack problem belongs to the domain of _____ problems.

Optimization (p91) NP Complete Linear Solution

Sorting

Question # 92

Suppose we have three items as shown in the following table, and suppose the capacity of the knapsack is 50 i.e.

$W=50$.

Item	Value	Weight
1	60	10
2	100	20
3	120	30

The optimal solution is to pick

Items 1 and 2

Items 1 and 3

Items 2 and 3

None of these

Question # 93

Theta asymptotic notation for $T(n)$:

Select correct option:

Set of functions described by: $c_1g(n) \leq f(n)$ for c_1 some constant and $n=n_0$

Set of functions described by $c_1g(n) \geq f(n)$ for c_1 some constant and $n=n_0$

Theta for $T(n)$ is actually upper and worst case complexity of the code

Set of functions described by: $c_1g(n) \leq f(n) \leq c_2g(n)$ for c_1 and c_2 some constants and $n=n_0$

Question # 94

A (an) _____ is a left-complete binary tree that conforms to the heap order.

Heap (page 40)

Binary tree

Binary search tree

Array

Question # 95

A heap is a left-complete binary tree that conforms to the _____

increasing order only

decreasing order only

heap order (p40)

(log n) order

Question # 96

Consider the following Algorithm: $\text{Fun}(n)\{ \text{if } (n=1) \text{ return } 1 \text{ else return } (n * \text{Fun}(n-1)) \}$ Recurrence for the above algorithm is: Select correct option:

$nT(n-1)+1$

$2T(n-1)+1$

$T(n-1)+cn$

$T(n-1)+1$

Question # 97

The recurrence relation of Tower of Hanoi is given below $T(n)=\{1 \text{ if } n=1 \text{ and } 2T(n-1) \text{ if } n > 1\}$ In order to move a tower of 5 rings from one peg to another, how many ring moves are required?

Select correct option:

16

10

32 (p30)

31

Question # 98

In Quick Sort Constants hidden in $T(n \log n)$ are

1. Large

2. Medium

3. Small (Reference)

4. Not Known

Question # 99

In stable sorting algorithm:

One array is used

In which duplicating elements are not handled.

More than one arrays are required.

Duplicating elements remain in same relative position after sorting. (p54)

Question # 100

In in-place sorting algorithm is one that uses arrays for storage:

An additional array

No additional array (p54)

Both of above may be true according to algorithm

More than 3 arrays of one dimension

Question # 101

Which may be a stable sort?

1. Merger 2. Insertion **3. Both above (p54)** 4. None of the above

Question # 102

An in place sorting algorithm is one that uses _____ arrays for storage

1. Two dimensional arrays 2. More than one array
3. No Additional Array (p54) 4. None of the above

Question # 103

Quick sort is _____

Stable and In place **Not stable but in place (p54)**

Stable and not in place Sometime in place and sometime stable

Question # 104

One Example of **in place** but **not stable** sort is

Quick (p54) Heap Merge Bubble

Question # 105

Which may be stable sort?

Bubble sort Insertion sort **both of above (p54)**

Question # 106

Merge sort is stable sort, but not an in-place algorithm **True (p54)** False

A sorting algorithm is **stable** if duplicate elements remain in the same relative position after sorting.

(p54)

Inplace (SBHIQ)	Stable (MCBI)
Selection Sort	Merge Sort
Bubble Sort	Counting Sort
Heap Sort	Bubble Sort
Insertion Sort	
Quick Sort	

Question # 107

$8n^2 + 2n - 3$ grows asymptotically at least as fast as **n^2** .

Question # 108

Given a set of points $P = \{p_1, p_2, \dots, p_n\}$ in 2-space, output the set of maximal points of P , i.e., those points p_i such that p_i is not dominated by any other point of P .

True (p11) False

Question # 109

While Sorting, the ordered domain means for any two input elements x and y _____ satisfies only.

$x < y$

$x > y$

$x = y$

All of the above (p34)

Question # 110

Brute-force algorithm for 2D-Maxima is operated by comparing _____ pairs of points.

Two Some Most **All (p18)**

Question # 111

Merge sort is based on _____

Brute-force Plan-sweep Axis-sweep **Divide and Conquer (p27)**

Question # 112

In the statement "if $(P[i].x < P[j].x) \& (P[i].y < P[j].y)$ ", the number of times elements of P are accessed is _____

1 2 3 **4 (p14)**

Question # 113

Al-Khawarizmi was a/an....

Artist **Mathematician** Astronomer Khalifah

Question # 114

In Sorting the key value or attribute _____ from an ordered domain.

Must be Not always



The key value need not be a number. It can be any object from a **totally** ordered domain. (p39)

Question # 115

In Selection problem, the Sieve technique works in ____

Non-recursive manner

Constant time

Phases (p34)

One complete go

Question # 116

In max heap (for Heap Sort algorithm), when every time maximum element is removed from top we replace it with ____ leaf in the tree.

Second last

Last (p41)

First

Any

Question # 117

If input "n" is odd, the median will be ____

$(n+1)/2$ (p34)

$n/2$

Question # 118

The function $f(n) = n(\log n + 1)/2$ is asymptotically equivalent to $n \log n$. Here **Upper Bound** means the function $f(n)$ grows asymptotically _____ faster than $n \log n$.

More

Quiet

Not (p23)

At least

Question # 119

In asymptotical analysis of $n(n - 3)$ and $4n*n$, as n becomes large, the dominant (fastest growing) term is some constant times ____.

$n+1$

$n-1$

n

$n*n$ (p23)

Question # 120

In asymptotical analysis of $n*(5 + 2)-3$, as n becomes large, the dominant (fastest growing) term is some constant times ____.

$n+1$

$n-1$

n

$n*n$

Question # 121

A point p in 2-dimensional space is usually given by its integer coordinate(s)_____

Select correct option:

$p.x$ only

$p.y$ only

$p.x$ & $p.z$

$p.x$ & $p.y$ (p10)

Question # 122

Algorithm analysts know for sure about efficient solutions for NP-complete problems.

True

False (p9)

Question # 123

In _____ we have to find rank of an element from given input.

Select correct option:

Merge sort algorithm

Selection problem (p34)

Brute force technique

Plane Sweep algorithm

Question # 124

In Heap Sort algorithm, if heap property is violated _____

Select correct option:

We call Build heap procedure

We call Heapify procedure (p41)

We ignore

Heap property can never be violated

Question # 125

Upper bound requires that there exist positive constants c_2 and n_0 such that $f(n) \leq c_2 n$ for all $n \leq n_0$

Select correct option:

Less than

Equal to or Less than (p25 => for all $n \geq n_0$)

Equal or Greater than

Greater than

Question # 126

A RAM is an idealized **algorithm** with takes an infinitely large random-access memory.

True

False (p10 => A RAM is an idealized **machine)**

Question # 127

In simple brute-force algorithm, we give no thought to efficiency.

True (p11)

False

Question # 128

_____ is one of the few problems, where provable lower bounds exist on how fast we can sort.

Select correct option:

Searching

Sorting (p39)

Both Searching & Sorting

Graphing

Question # 129

The total no of arguments passed to merge sort algorithm is _____

2

3 (p28 => Array itself + two indices)

4

5

Question # 130

While solving Selection problem, in Sieve technique we partition input data _____

In increasing order

In decreasing order

According to Pivot (p35)

Randomly

Question # 131

The number of accesses to any element of space is **not counted** for the running time calculation of algorithm.

True

False (p14)

Question # 132

In merge sort algorithm, to merge two lists of size $n/2$ to a list of size n , takes _____ time.

Theta (n) (p30)

Theta $\log(n)$

Theta $\log_2(n)$

Theta $n\log(n)$

Question # 133

In Heap Sort algorithm, Heapify procedure is _____ in nature.

Recursive (p43) Non-Recursive

Question # 134

In Heap Sort algorithm, we call Build-heap procedure _____

Only once Twice Thrice **As many times as we need**

Question # 135

In merge sort algorithm, we split the array around the _____ index q.

Entring **Mid (p28)** Exiting Summing

Question # 136

The Iteration method is used for _____

Comparing sorting algorithms only

Solving Recurrence relations (p31)

Merging elements in Merge sort

Dividing elements in Merge sort

Question # 137

Rank of an element can be defined as _____

One minus the number of elements that are smaller

Two plus the number of elements that are greater

One plus the number of elements that are smaller (p34)

Two minus the number of elements that are smaller

Question # 138

$f(n)$ is a set of functions such that there exist positive constants c_1 , c_2 and n_0 such that $0 \leq c_1g(n) \leq f(n) \leq c_2g(n)$ for all $n \geq n_0$. Then $f(n)$ is asymptotically _____ $g(n)$.

Select correct option:

Less than

Greater than

Equivalent to (p23)

Not equivalent to

Question # 139

Floor and ceiling are _____ to calculate while analyzing algorithms.

Select correct option:

Very easy

Usually considered difficult

(p31)

Question # 140 In Heap Sort algorithm, the maximum levels an element can move upward is _____

Select correct option:

Theta (log n)

(p43)

Order (log n)

Omega (log n)

O (1) i.e.

Constant time

Question # 141

In Heap Sort algorithm, the total running time for Heapify procedure is _____

Select correct option:

Theta (log n)

Order (log n)

(p43)

Omega (log n)

O (1) i.e. Constant time

Question # 142

In pseudo code, the level of details depends on intended audience of the algorithm.

True (p12)

False

Question # 143

Counting sort assumes that the numbers to be sorted are in the range _____

CS502 Solved Grand Quiz 2020

Question No 1

In the analysis of algorithms, _____ plays an important role.

Select Correct Option

- ▶ Text analysis
- ▶ **Time**
- ▶ Growth rate
- ▶ Money

Question No 2

_____ overcomes the limitations of _____ by working as per positional notations of numbers.

Select Correct Option

- ▶ **Bubble sort, Radix sort**.....(For More Visit VUStudents.pk)
- ▶ Counting sort, Radix sort
- ▶ Radix sort, Bubble sort
- ▶ Radix sort, Counting sort

Question No 3

In generating Fibonacci Sequence, we can avoid unnecessary repetitions by _____ process.

Select Correct Option

- ▶ Tokenization
- ▶ **Memoization**.....([For More Visit VUStudents.pk](#))
- ▶ Randomization
- ▶ Memorization

Question No 4

In Dynamic Programming, our approach is to _____.

Select Correct Option

- ▶ **Develop the solution in a top-down fashion**
- ▶ Express the problem non-recursively
- ▶ Build the solution in a bottom-up fashion
- ▶ Input several sub-problems simultaneously

For More Visit [VUStudents.pk](#)

Question No 5

We can use the optimal substructure property to devise a _____ formulation of the edit distance problem.

Select Correct Option

- ▶ Selective
- ▶ Optimum
- ▶ Iterative
- ▶ **Recursive.....(For More Visit VUStudents.pk)**

Question No 6

Chain matrix multiplication problem can be solved through _____ strategy.

Select Correct Option

- ▶ **Dynamic programming.....(For More Visit VUStudents.pk)**
- ▶ Greedy
- ▶ Divide and conquer
- ▶ Implementing arrays

Question No 7

For 2D Maxima problem, Plane Sweep algorithm first of all _____-.

Select Correct Option

- ▶ **Sorts all points.....(For More Visit VUStudents.pk)**
- ▶ Delete some points
- ▶ Output the elements
- ▶ Pushes all points on stack

Question No 8

_____ algorithm based on Dynamic Programming strategy.

Select Correct Option

- ▶ **Heap Sort.....**(For More Visit VUStudents.pk)
- ▶ Binary Tree
- ▶ Quick Sort
- ▶ Edit distance

For More Visit VUStudents.pk

Question No 9

The only way to convert a string of i characters into the empty string is with i deletions, represented as _____--.

Select Correct Option

- ▶ $E(0, j) = j$
- ▶ $E(i, j) = 1$
- ▶ $E(0, i) = 1$
- ▶ **$E(i, 0) = i$(For More Visit VUStudents.pk)**

Question No 10

Edit Distance algorithm based on _____ strategy.

Select Correct Option

- ▶ Greedy
- ▶ **Dynamic Programming.....(For More Visit VUStudents.pk)**
- ▶ Divide and Conquer
- ▶ Brute force

Question No 11

In _____ we have to find rank of an element from given input.

Select Correct Option

- 5 Merge sort algorithm
- 6 **Selection problem**.....(For More Visit VUStudents.pk)
- 7 Brute force technique
- 8 Plane Sweep algorithm

Question No 12

In the statement “output P[i].x, P[i].y”, the number of time elements of P are accessed is _____

Select Correct Option

- P 1
- Q 2.....(For More Visit VUStudents.pk)
- R 3
- S 4

For More Visit VUStudents.pk

Question No 13

Rank of an element can be defined as _____-.

Select Correct Option

- ✓ One minus the number of elements that are smaller
- ✓ Two plus the number of elements that are greater
- ✓ **One plus the number of elements that are smaller**
- ✓ Two minus the number of elements that are smaller

Question No 14

$8n^2 + 3n - 3$ will eventually exceed $c^2 \cdot (n)$ no matter how large we make _____-.

Select Correct Option

- ✓ n
- ✓ c^2
- ✓ **$2n$(For More Visit VUStudents.pk)**
- ✓ this equation

Question No 15

Matrix Multiplication is a(n) _____ operation.

Select Correct Option

- ✓ Commutative
- ✓ **Associative.....(For More Visit VUStudents.pk)**
- ✓ Neither commutative nor associative
- ✓ Transitive

Question No 16

The number of nodes in a complete binary tree of height h is:

Select Correct Option

- ✓ $2 * (h+1) - 1$
- ✓ **$2^{(h+1)} - 1$** (For More Visit VUStudents.pk)
- ✓ $2^{*(h+1)}$
- ✓ $((h+1) ^ 2) - 1$

Question No 17

Applying the sieve technique to selection problem, _____ element is picked from array.

Select Correct Option

- ✓ Output
- ✓ Total
- ✓ Input
- ✓ **Pivot.....**(For More Visit VUStudents.pk)

Question No 18

In Quick sort algorithm, we choose pivot _____--.

Select Correct Option

- ✓ **Always the smallest element**(For More Visit VUStudents.pk)
- ✓ Greater than 5
- ✓ Randomly
- ✓ Less than 5

Question No 19

Approach of solving geometric problems by sweeping a line across the plane is called _____ sweep.

Select Correct Option

- ✓ Line
- ✓ **Plane.....**(For More Visit VUStudents.pk)
- ✓ Cube
- ✓ Box

Question No 20

In Knapsack Problem, the goal is to put items in the knapsack such that the value of the items is _____ subject to weight limit of knapsack.

Select Correct Option

- ✓ Minimized
- ✓ Decrease
- ✓ **Maximized.....**(For More Visit VUStudents.pk)
- ✓ None of the given options

For More Visit VUStudents.pk

Question No 21

Sorting is performed on the basis of _____.

Select Correct Option

- ✓ Computational resources
- ✓ Asymptotic notation
- ✓ Summation
- ✓ **Some key value or attribute**.....([For More Visit VUStudents.pk](http://VUStudents.pk))

Question No 22

In Asymptotical analysis of $n(n-3)$ and $4n*n$, as n becomes large, the dominant (fastest growing) term is some constant times_____.

Select Correct Option

- ✓ $n+1$
- ✓ $n-1$
- ✓ n
- ✓ **$n*n$**([For More Visit VUStudents.pk](http://VUStudents.pk))

Question No 23

In Heap Sort algorithm, we build _____ for ascending sort.

Select Correct Option

- ✓ Max heap
- ✓ **Min heap**.....([For More Visit VUStudents.pk](http://VUStudents.pk))
- ✓ Full heap
- ✓ Order heap

Question No 24

If we have an equation $8n^2 + 7f \cdot n + 5f + 6$ then n is large, _____ term will be much larger than the n term and will dominate the running time.

Select Correct Option

- ✓ $f(g(n))$
- ✓ $g(n)^2$
- ✓ **n^2** (For More Visit VUStudents.pk)
- ✓ $f(n)$

Question No 25

_____ is the process of avoiding unnecessary repetitions by writing down the results of recursive calls and looking them up again if we need them later.

Select Correct Option

- ✓ Loop
- ✓ **Memoization.....**(For More Visit VUStudents.pk)
- ✓ Recursion
- ✓ Function

Question No 26

The worst-case running time of merge sort is _____ in order to sort an array of n elements.

Select Correct Option

- ✓ $O(\log n)$
- ✓ $O(n)$
- ✓ **$O(n \log n)$ **(For More Visit VUStudents.pk)
- ✓ $O(n^2)$

Question No 27

If matrix A of dimension 2×4 is multiplied with matrix b of dimension 4×3 , then the dimension of resultant matrix will be _____.

Select Correct Option

- ✓ **2×4 **(For More Visit VUStudents.pk)

- ✓ 4 x 3
- ✓ 3 x 4
- ✓ 2 x 3

Question No 28

Fibonacci Sequence was named on _____ , a famous mathematician in 12th Century.

Select Correct Option

- ✓ Fred Brooks
- ✓ Grady Booch
- ✓ **Leonardo Pisano**.....([For More Visit VUStudents.pk](#))
- ✓ Edger F.Codd

For More Visit [VUStudents.pk](#)

Question No 29

Algorithm is a sequence of computational steps that _____ the input into output.

Select Correct Option

- ✓ Merge
- ✓ Assign
- ✓ **Transform** (For More Visit VUStudents.pk)
- ✓ Integrate

Question No 30

If the indices passed to merge sort algorithm are _____, then this means that there is only one element to sort.

Select Correct Option

- ✓ Small
- ✓ Large
- ✓ **Equal**
- ✓ Not Equal

Question No 1

The tricky part of _____ algorithm(s) is/are, how to detect whether the addition of an edge will create a cycle in viable set A.

Kruskal's (Page 147)

Question No 2

Overall Running time of Prim's algorithm is _____.

$\Theta(E \log V)$ (Page 153)

Question No 3

Finding the faster result of the shortest path from u to v for every pair of vertices u and v we use_____.

All-pairs shortest-paths problem (Page 154)

Question No 4

What is the time complexity to extract a vertex from the priority queue in Prim's algorithm?

$O(\log V)$ (Page 152)

Question No 5

Edge weights can be interpreted as distance _____.

in the shortest-paths (Page 153)

Question No 6

Dijkstra's Algorithm is used to solve_____problems.

Single-source shortest path (Page 159)

Question No 7

Problems such as the shortest route between cities can be solved efficiently by modeling the road map as a_____.

Graph (Page 153)

Question No 8

For _____ graphs, there is no distinction between forward and back edges.

Undirected (Page 130)

Question No 9

Bellman-Ford Algorithm does not allow $G(\text{graph})$ to have _____.

negative cost cycles (Page 159)

Question No 10

A topological sort of a DAG is a _____ ordering of the vertices of the DAG such that for each edge (u, v) , u appears before v in the ordering.

Linear (Page 133)

Quiz File 2

Question No 1

Networks are _____ in the sense that it is possible from any location in the network to reach any other location in the digraph.

Complete (Page 135)

Question No 2

According to parenthesis lemma vertex u is a descendent of v vertex if and only if:

$f[d[u], f[u]] \leq [d[v], f[v]]$ (Page 129)

Question No 3

For _____ graphs, there is no distinction between forward and back edges.

Undirected (Page 130)

Question No 4

There exist a unique path between any _____ vertices of a free tree.

Two (Page 142)

Question No 5

Keeping in mind the shortest-path, if given scenarios occur in computer networks like the internet where data packets have to be routed. The vertices are_____and Edges are _____which may be wired or wireless.

Routers, communication links (Page 153)

Question No 6

In undirected graph, by convention all the edges are called _____ edges.

back (Page 130)

Question No 7

Bellman-Ford algorithm is slower than_____.

Dijkstra's (Page 159)

Question No 8

From the given option's which one is correct regarding the time complexity of Dijkstra's algorithm:

$O(N^2)$

Question No 9

In the shortest-paths problem, we are given a weighted of_____ $G = (V, E)$.

Un-directed graph

Question No 10

You have an adjacency list for G , what is the time complexity to compute Graph transpose G^T ?

$(V+E)$

CS502 – Midterm Spring 2013 (26 May-05 June)

How would you modify QUICKSORT to sort into non-increasing order?

Answer: - [Click here for detail](#)

To make QUICKSORT sort in non-increasing order we must modify PARTITION.

Q- We can avoid unnecessary repetitions for recursive calls?

Answer: - (Page 74)

We can avoid these unnecessary repetitions by writing down the results of recursive calls and looking them up again if we need them later. This process is called *memoization*.

Q-Write a pseudo code Fibonacci With memorization? -- (3)

Answer: - (Page 74)

MEMOFIB(n)

1 **if** (n < 2)

2 **then return** n

3 **if** (F[n] is undefined)

4 **then** F[n] = MEMOFIB(n - 1) + MEMOFIB(n - 2)

5 **return** F[n]

Spelling correction in edit distance? 3 marks

Answer: - (Page 76)

If a text contains a word that is not in the dictionary, a 'close' word, i.e. one with a small edit distance, may be suggested as a correction. Most word processing applications, such as Microsoft Word, have spelling checking and correction facility. When Word, for example, finds an incorrectly spelled word, it makes suggestions of possible replacements.

Bubble sort?

Answer: - (Page 39)

Scan the array. Whenever two consecutive items are found that are out of order, swap them. Repeat until all consecutive items are in order.

What is the worst case running time for the Quick sort? What simple change is required in the algorithm to preserve its linear expected running time and makes it worst case time $\Theta(n \log n)$

Answer: - (Page 49)

Worst case running time is $O(n^2)$.

The simple change which can change the running time of the edit distance algorithm is the number of entries n^2 .

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QUICK, HEAP, COUNTING, MERG KA TABLE KA JIS ME STABLE OR IN-PLACE BTANEY THE 5 MARKS

Answer: Page 54

IN Place	Stable
Selection Sort	
Quick Sort	
Heap sort	

Bubble sort	Bubble sort
Insertion Sort	Insertion Sort
	Merge sort
	Counting sort

ADIT DISTANCE KI 3 APPLICATION K NAME BTANEY THE 3 MARKS

Answer: - (Page 76)

Spelling Correction
Plagiarism Detection
Speech Recognition

MEMOIZATION KI DIFINATION THI 2 MARKS

Answer: - [Click here for detail](#)

Memoization is an optimization technique used primarily to speed up computer programs by having function calls avoid repeating the calculation of results for previously processed inputs

WORS CASE OR AVERAGE CASE DEFINE KRNE THE 2 MARKS

Answer: - (Page 13)

Worst-case time is the maximum running time over all (legal) inputs of size n

Average-case time is the average running time over all inputs of size n .

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Write Essential constraint of Counting sort.

Answer:

Essential constraint for the counting sort is that numbers to be sorted must be small integers i.e., $1 \dots k$ where k is small.

What are total numbers of entries in matrix for edit distance?

Answer: - (Page 84)

There are $O(n^2)$ entries in the matrix. Each entry $E(i, j)$ takes $O(1)$ time to compute. The total running time is $O(n^2)$.

WHAT IS the necessary assumption for average case analysis quick sort?

Answer: - (Page 50)

Average case depends on some assumption about the distribution of inputs. However, in the case of quicksort, the analysis does not depend on the distribution of input at all. It only depends upon the random choices of pivots that the algorithm makes

Suggest & describe modifications of the implementation of "Quick sort" that will improve its performance?

Answer: - (Page 76)

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Q1 : How we Heapify?

Answer:

If an element in the Heap is not at its proper place means it is violating the Heap Order, the Heapify procedure is used to fix it and place it at its proper position. In Heapify, we recursively swap the element with its larger one child and stop at a stage when this element is larger than both of its children or it becomes the leaf node.

Q2 : Describe Heap sort algorithm?

Answer: - (Page 43)

HEAPIFY(array A, int i, int m)

1 l LEFT(i)

2 r RIGHT(i)

3 max i

4 if (l < m) and (A[l] > A[max])

5 then max = l

6 if (r < m) and (A[r] > A[max])

7 then max = r

8 if (max ≠ i)

9 then SWAP(A[i], A[max])

10 HEAPIFY(A, max, m)

Q3 : How many elements are in matrix of edit distance?

Answer: Rep

Q6 : suggest and describe one modification of implementing quick sort? 5marks

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Edit Distance in Speech Recognition

Answer: - (Page 77)

Speech Recognition

Algorithms similar to those for the edit-distance problem are used in some speech recognition systems. Find a close match between a new utterance and one in a library of classified utterances.

What is sorting? Describe slow running sorting algorithms. 5 marks

Answer: - (Page 39)

Sorting is any process of arranging items in some sequence and/or in different sets, and accordingly,

There are a number of well-known slow $O(n^2)$ sorting algorithms. These include the following:

Bubble sort: Scan the array. Whenever two consecutive items are found that are out of order, swap them. Repeat until all consecutive items are in order.

Insertion sort: Assume that $A[1..i - 1]$ have already been sorted. Insert $A[i]$ into its proper position in this sub array. Create this position by shifting all larger elements to the right.

Selection sort: Assume that $A[1..i - 1]$ contain the $i - 1$ smallest elements in sorted order. Find the smallest element in $A[i..n]$ Swap it with $A[i]$. These algorithms are easy to implement. But they run in $_ (n^2)$ time in the worst case

Catalan Numbers and their formula 3 marks

Answer: - (Page 85)

Catalan numbers are related the number of different binary trees on n nodes. Catalan number is given by the formula:

$$C(n) = \frac{1}{n+1} \binom{2n}{n}$$

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Q1.Heap Sort , Counting Sort , Quick Sort ,Merge Sort , in ,table ,And required to tell , which is inplace Algorithm and stable ? Marks 5

Answer: rep

Q2 . Suggest that how can make better improvement in Quick Sort algorithm .Marks 5

Answer: - (Page 54)

The best we have seen so far is $O(n \log n)$ algorithms for sorting. Is it possible to do better than $O(n \log n)$? If a sorting algorithm is solely based on comparison of keys in the array then it is *impossible* to sort more efficiently than $(n \log n)$ time.

Q3. Consider three numbers with comparison based sortig algorithm and write possible combination in a1,a2,a3 .Marks 3

Answer: - (Page 85)

Consider sorting three numbers a_1, a_2, a_3 . There are $3! = 6$ possible combinations:

$(a_1, a_2, a_3), (a_1, a_3, a_2), (a_3, a_2, a_1)$

$(a_3, a_1, a_2), (a_2, a_1, a_3), (a_2, a_3, a_1)$

Q4. What is better aproach of multiplication rather than straight form of Multiplication . Named that . Marks 3

Answer: - (Page 54)

Chain Matrix Multiplication-Dynamic Programming Formulation

Q6.worst case and average case algorithm of Quick Sort .Marks 2

Answer: - (Page 50)

Answer: Worst case = $O(n^2)$

Average Case = $O(n \log n)$

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1.worst case of bucket algorithm

Answer: - $O(n)$

Catalan Numbers and their formula 3 marks

Answer: rep

speech reorganization.2

Answer: rep

Worst and average case of quick sort algorithm.2 marks

Answer: rep

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Radix sort table ? 5 marks

Answer:

Radix sort:

In linear time sorting, counting sort is used for the numbers in the range 1 to k where k is small and it is based on determining the rank of each number in final sorted array. But it is useful only for small integers i.e., $1 \dots k$ where k is small. But if k were very large, then the size of the rank array formed would also be very large which is not efficient. So solution for such cases is the Radix sort which works by sorting one digit at a time.

Example:

841 84[1] 8[4]1 [1]85

185 è 37[3] è 3[7]3 è [3]73

373 18[5] 1[8]5 [8]41

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1. What are the worst and average run times of quick sort? (2 marks)

Answer: rep

2. What are total numbers of entries in matrix for edit distance? (2 marks)

Answer: rep

Q.3 What does dynamic programming have in common with divide and conquer ?

Ans. : Both the divide and conquer and dynamic programming solve the problem by breaking it into number of subproblems. In both these methods solutions from subproblems are collected together to form a solution to given problem.

Q.4 What is the principle difference between the two techniques ?

Ans. : Both divide and conquer as well as dynamic programming solve the given problem by breaking it into subproblems. But in divide and conquer the subproblems are solved independently whereas in dynamic programming the subproblems share the solutions among themselves. The dynamic programming works on principle of optimality to solve the problem.

5. Show that the running time of quick sort when the array A contains distinct elements and sorted in decreasing order ? (5 marks)

Answer:

3. (CLRS 7.2-3) Show that the running time of QUICKSORT is $\Theta(n^2)$ when the array A contains distinct elements and is sorted in decreasing order.

Solution: On the first iteration of PARTITION the pivot element is chosen as the first element of A . Index i is incremented once and j is decremented until it reaches the pivot, i.e. the entire length of A . PARTITION returns to QUICKSORT the first element of A , which recursively sorts one subarray of size 1 and one of size $n - 1$. This process is repeated for the subarray of size $n - 1$. The running time of the entire computation is then given by the recurrence:

$$\begin{aligned} T(n) &= \begin{cases} \Theta(1) & n \leq 2 \\ T(n-1) + \Theta(n) & \text{otherwise} \end{cases} \\ &= \Theta(n^2). \end{aligned}$$

6. Illustrate how Radix Sort works on the following words: Show the result of first 2 passes only.

COW, DOW, SEA, RUG, ROW, MOB, BOX, TAB, BAR, EAR, TAR, DIG, BIG, TEA, NOW, FOX

Answer:

COW DOG SEA RUG ROW MOB BOX TAB BAR EAR TAR DIG BIG TEA NOW FOX
SEA TAB TEA BAR MOB EAR TAB TAR DOG SEA RUG TEA DIG DIG BIG BIG BAR
MOB EAR DOG TAR COW COW ROW ROW NOW NOW BOX BOX FOX FOX RUG BAR
BIG...

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Q: define heap and heap order

A *heap* is a left-complete binary tree that conforms to the *heap order*.

Q: Average-case Analysis of Quicksort

Average case analysis of Quick sort: The running time of Quick sort depends on the selection of pivot q which is done randomly. For average case analysis of quick sort, average is computed over all possible random choices of the pivot index q . The average case running time for quick sort is $\Theta(n \log n)$.

Q: Worst case Analysis of Quick sort

Worst case analysis of Quick sort: For worst case we maximize over all possible values of q , means the selection of pivot q which gives the maximum (worst) time for sorting. The chances are that the pivot values $q=1$ (start) or n (end) or $n/2$ (middle) happen to give maximum time values. The worst case running time is $O(n^2)$.

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In Random Access Machine, instructions are executed _____ Instructions are executed

Answer:- one-by-one (there is no parallelism). (Page 10)

Dynamic Programming algorithm:

Answer: - The genius of the algorithm is in the clever recursive formulation of the shortest path problem. (Page 162)

For the heap sort, access to nodes involves simple __ **arithmetic** __ operations. (Page 40)

Comparison based sorting algorithm cannot run faster than __ **$\Omega(n \log n)$** __ (Page 46)

Fibonacci sequence was posted by __ **Leonardo Pisano**, __ (Page 73)

Mid

Quiz by Attiq Kundi

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